KING SALMAN INTERNATIONAL UNIVERSITY

Faculty of Medicine

Histology

Lecture -1: Microscopy + Cytology-I

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2026/2027

Objectives

- **Compare** between light and electron microscopes
- **Classify** cytoplasmic organelles
- **Describe** the molecular structure of the cell membrane
- **Enumerate** the percentage of each component of cell membrane
- **Identify** the structure and function of cell coat
- **Corelate** between the structure and function of mitochondria

Histology



- What is Histology?

- *Microscopic* *Structure* of normal tissue

Microscopy



Measurements & Units

- Centimeter (cm) = 10 Millimeter (mm)
- Millimeter (mm) = 1000 Micrometer (μm)
- Micrometer (μm) = 1000 Nanometer (nm)

Types of Microscopy

Light Microscope



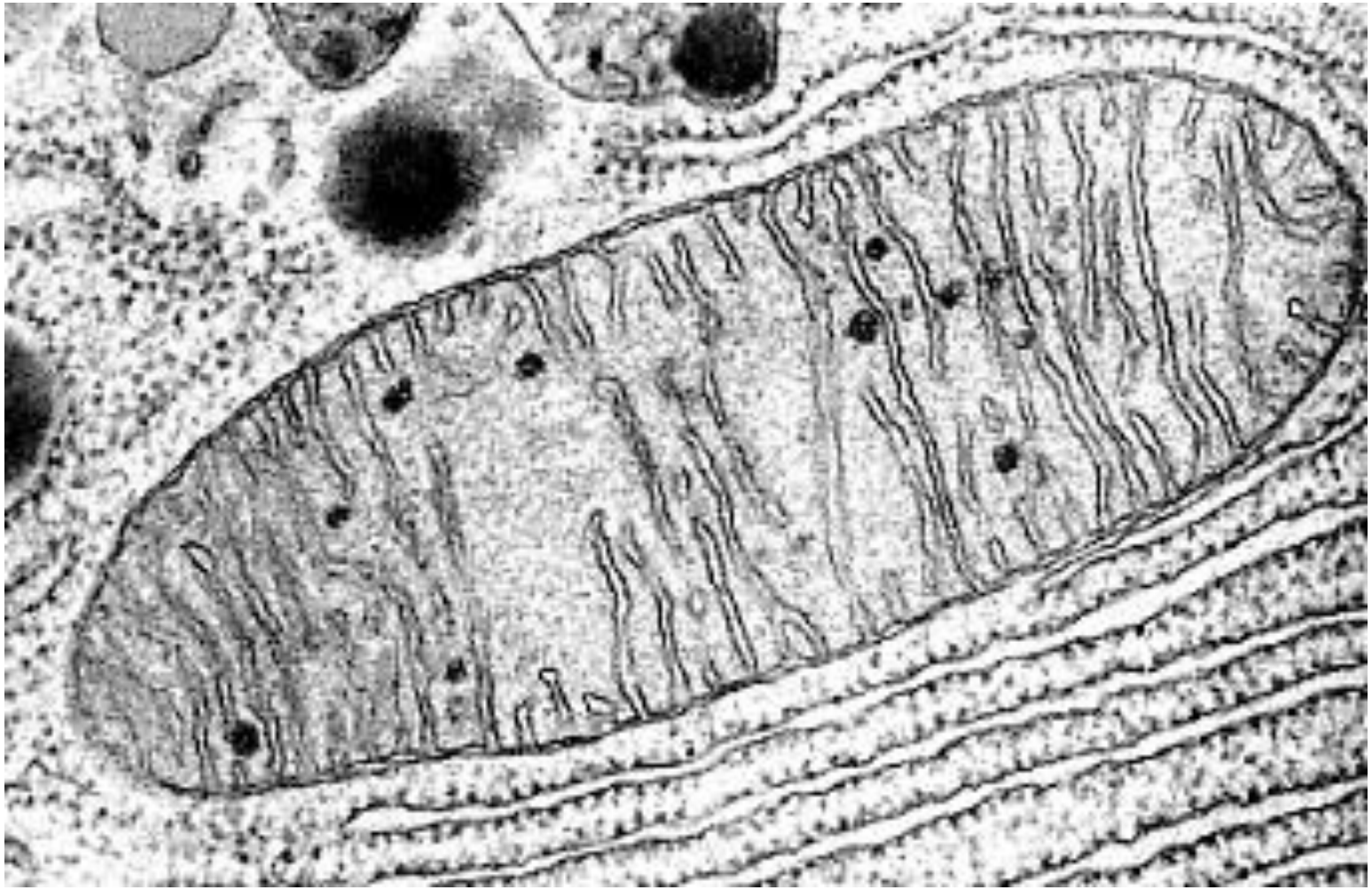
Electron Microscope



Difference between L.M & E.M

	Electron Microscope (E.M)	Light Microscope (L.M)
Source of illumination	electrons beam	light rays
Magnifiying power	x 10.000 & MUCH larger	up to x 1500-2000
Used for examination of	intracellular organelles	whole cells & tissues
Specimen treated with	Metal powder	H&E [Hematoxylin + Eosin] stain mostly
Types	Transmission / Scanning	only one type

Transmission - E.M

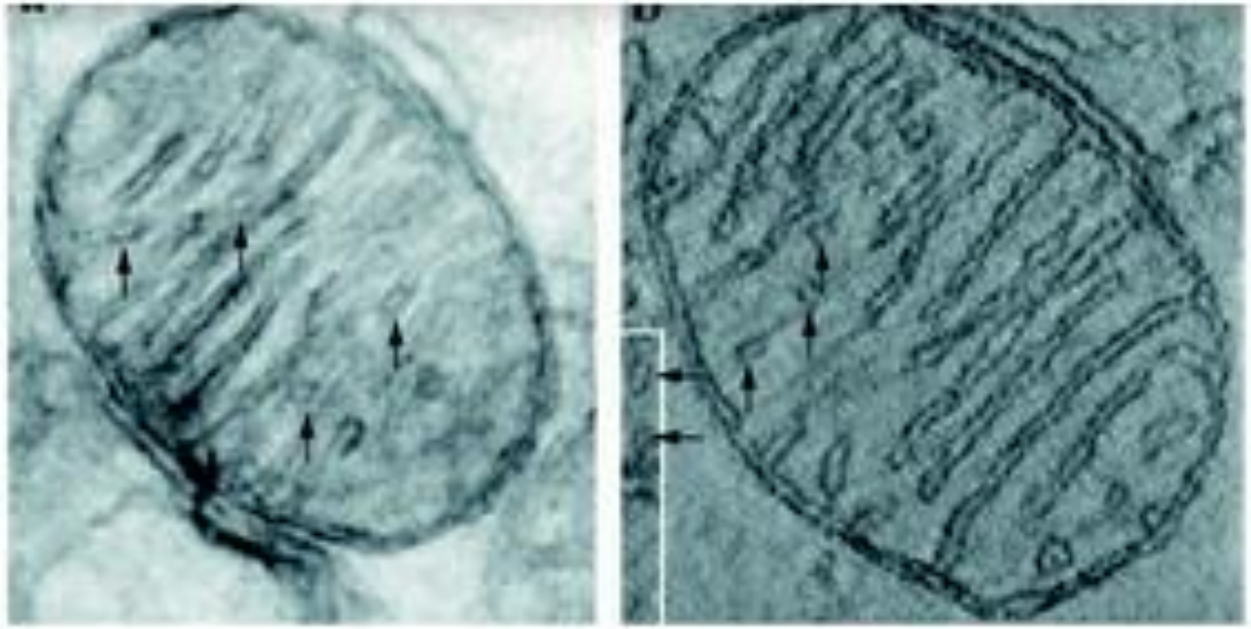


Scanning - E.M



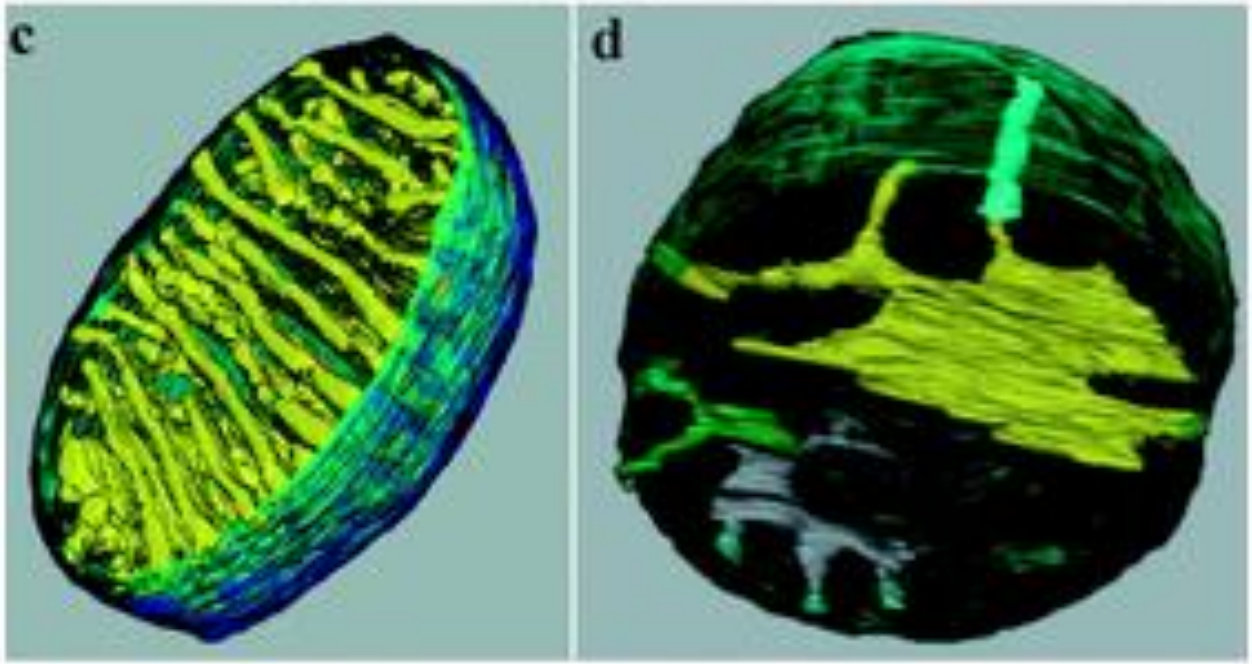
Transmission EM

pictures of mitochondria



Scanning EM

pictures of mitochondria

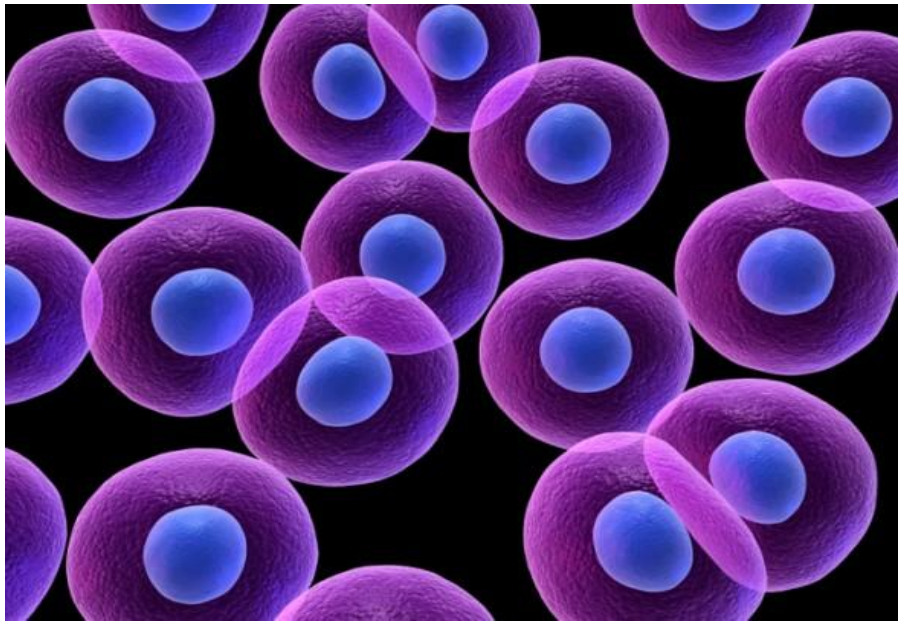


Mitochondria, as such, can **not** be **seen** with light microscopy. But their **site can be highlighted** with **special stains**. The way to visualize mitochondria directly is only through electron microscopy.


Cells
Tissues
Organs
Systems
Human Body

Cytology [The Cell]

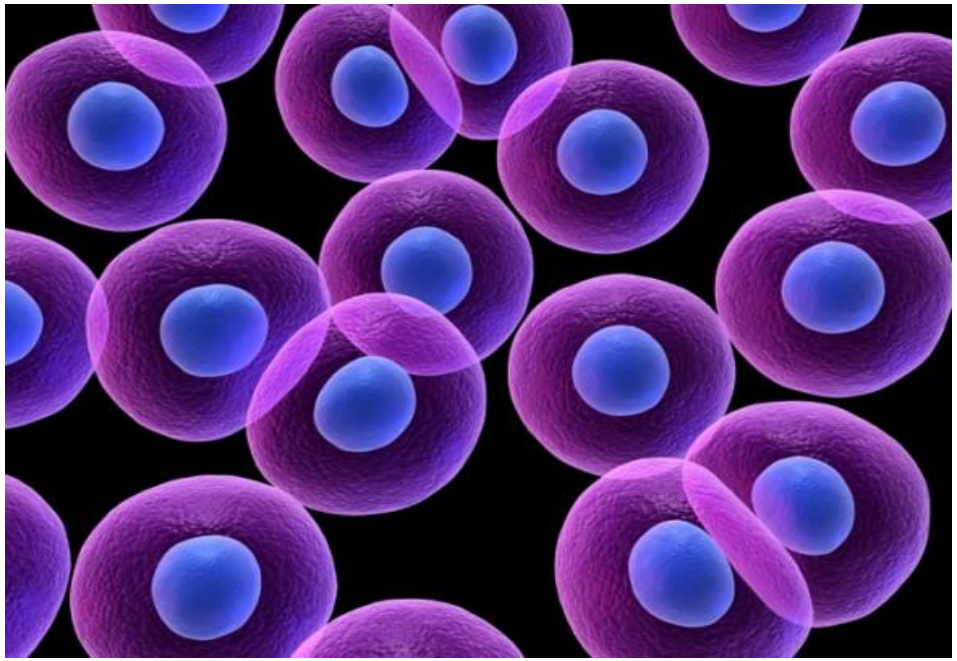
- Structural & functional unit
- Different shapes & sizes



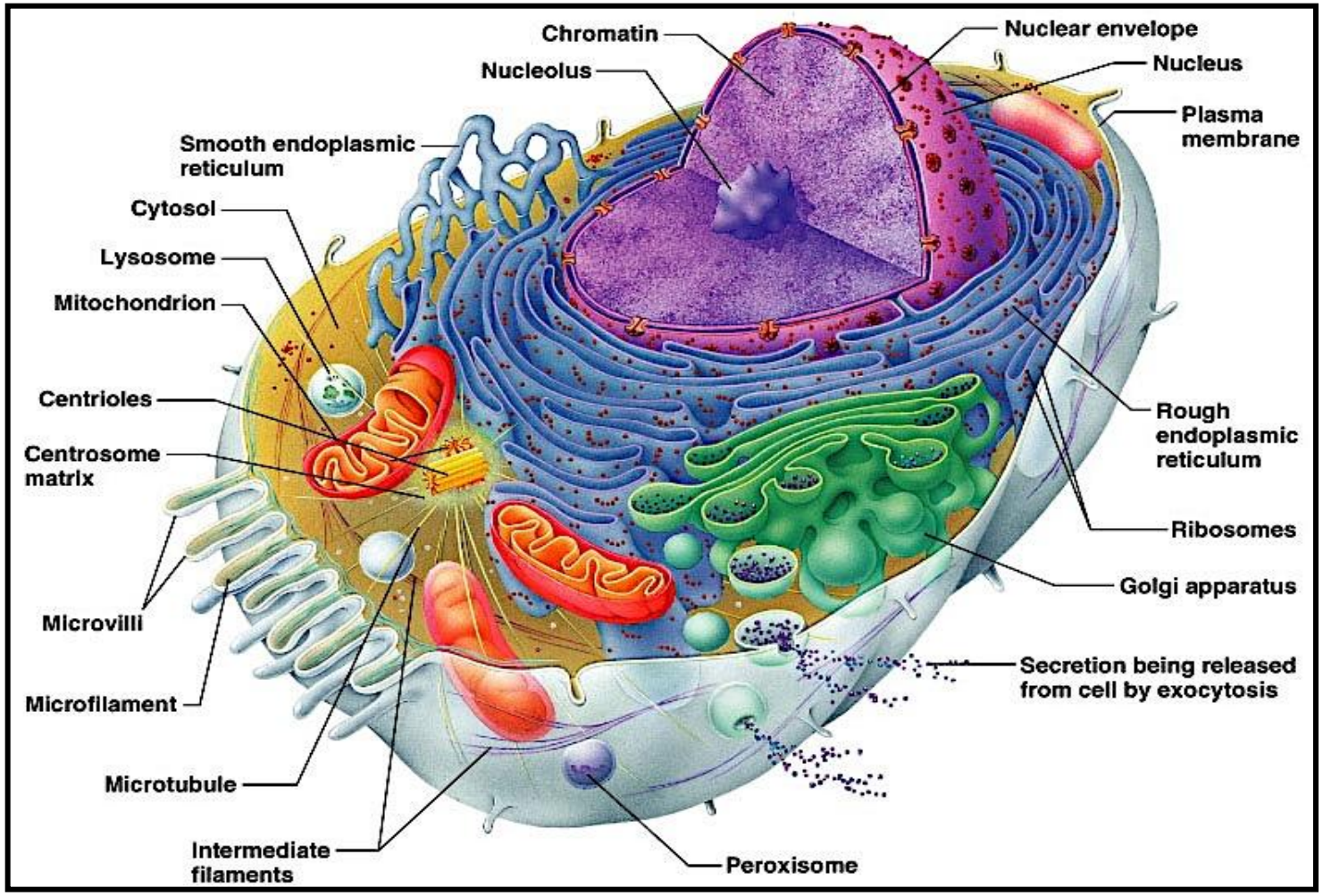
The Cell

Cytoplasm

Nucleus & organelles

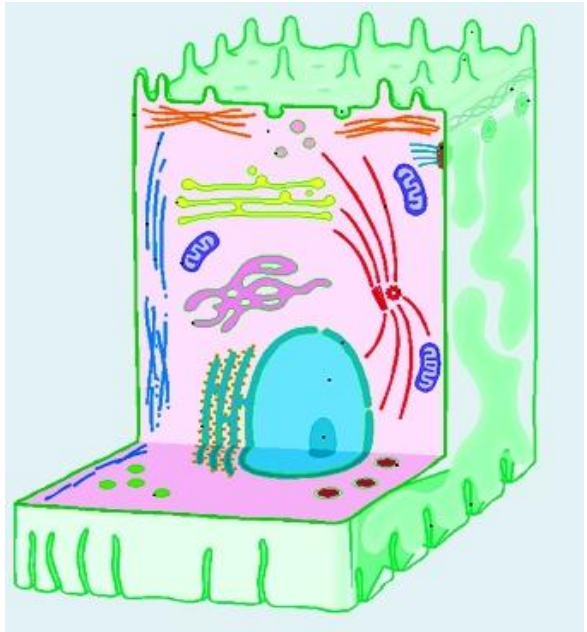


The living cell



Cytoplasm

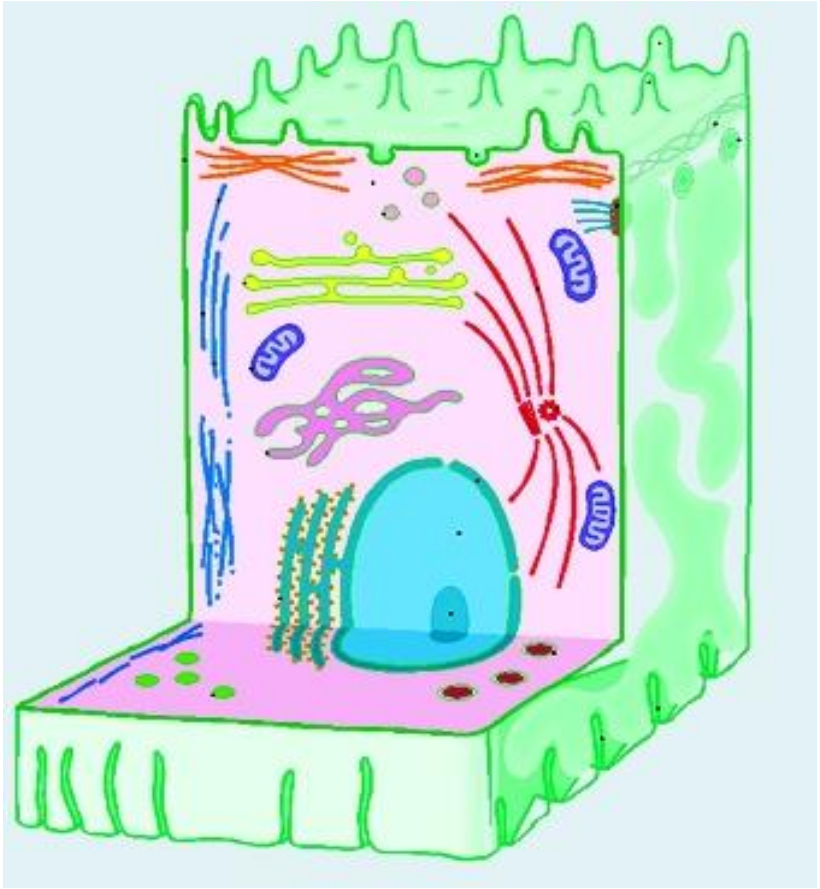
- Cytoplasmic Matrix
- Cytoplasmic organelles
- Cytoplasmic inclusions



1-Cytoplasmic Matrix

■**Colloid solution contains:**

- ✓ **Proteins**
- ✓ **Carbohydrates**
- ✓ **Lipids**
- ✓ **Minerals**
- ✓ **Enzymes**



2-Cytoplasmic organelles

- Living
- Permanent minute structures
- Essential for vital processes as respiration, secretion, digestion,….
- e.g.: mitochondria, Golgi apparatus, lysosomes,……..

3-Cytoplasmic inclusions

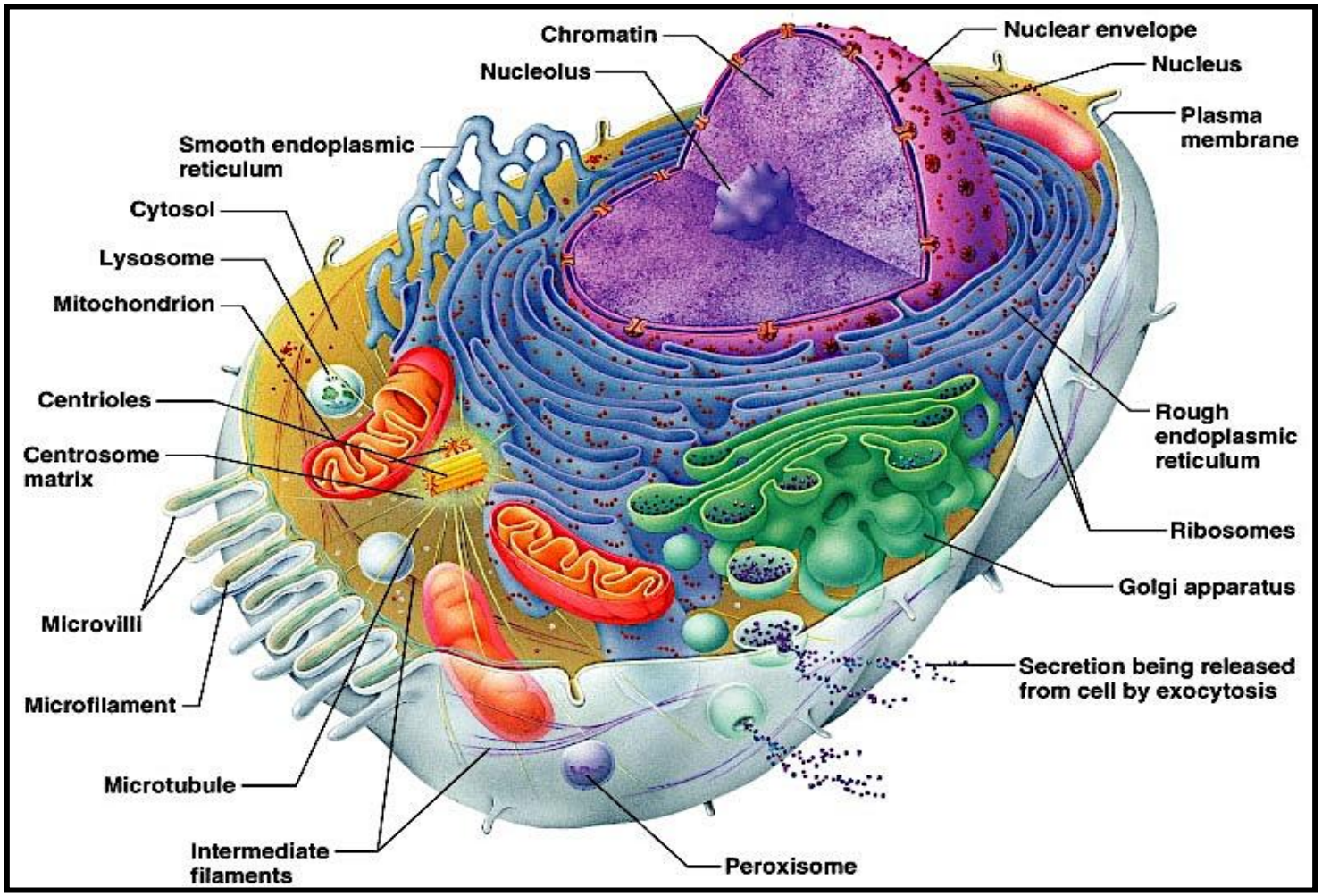
- Non-living
- Temporary structures
- Not essential for cell vitality
- e.g.: stored substances in some cells [Glycogen, Fat & Pigments]

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Types of Cytoplasmic Organelles

Membranous	Non-membranous
✓ Cell membrane	✓ Ribosomes
✓ Mitochondria	✓ Cytoskeleton: a-Microtubules (Centrioles & Cilia) b-Filaments (Thin, Intermediate & Thick)
✓ Endoplasmic reticulum (rough & smooth)	
✓ Golgi apparatus	
✓ Lysosomes	

The animal cell



1- Cell Membrane

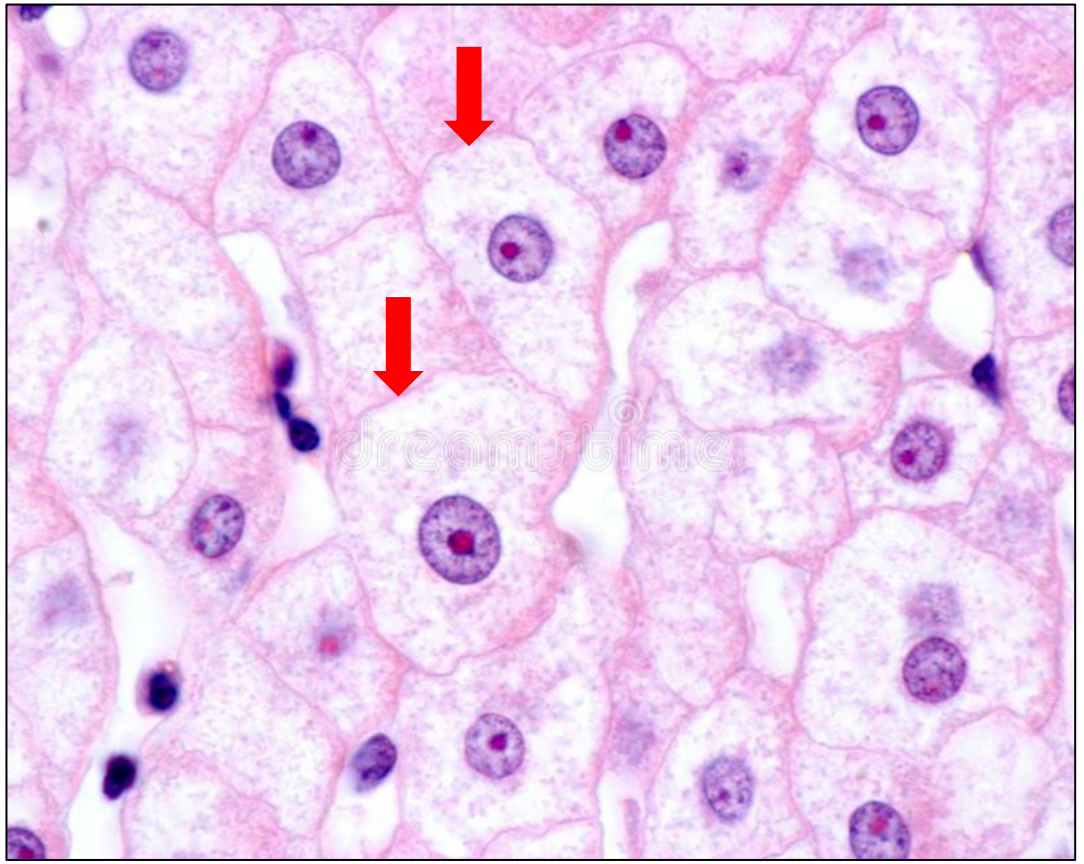
■**L.M:**

✓**Invisible as it is very thin**

[8-10 nm]

✓**But can be stained by PAS or**

Ag [Special stains]

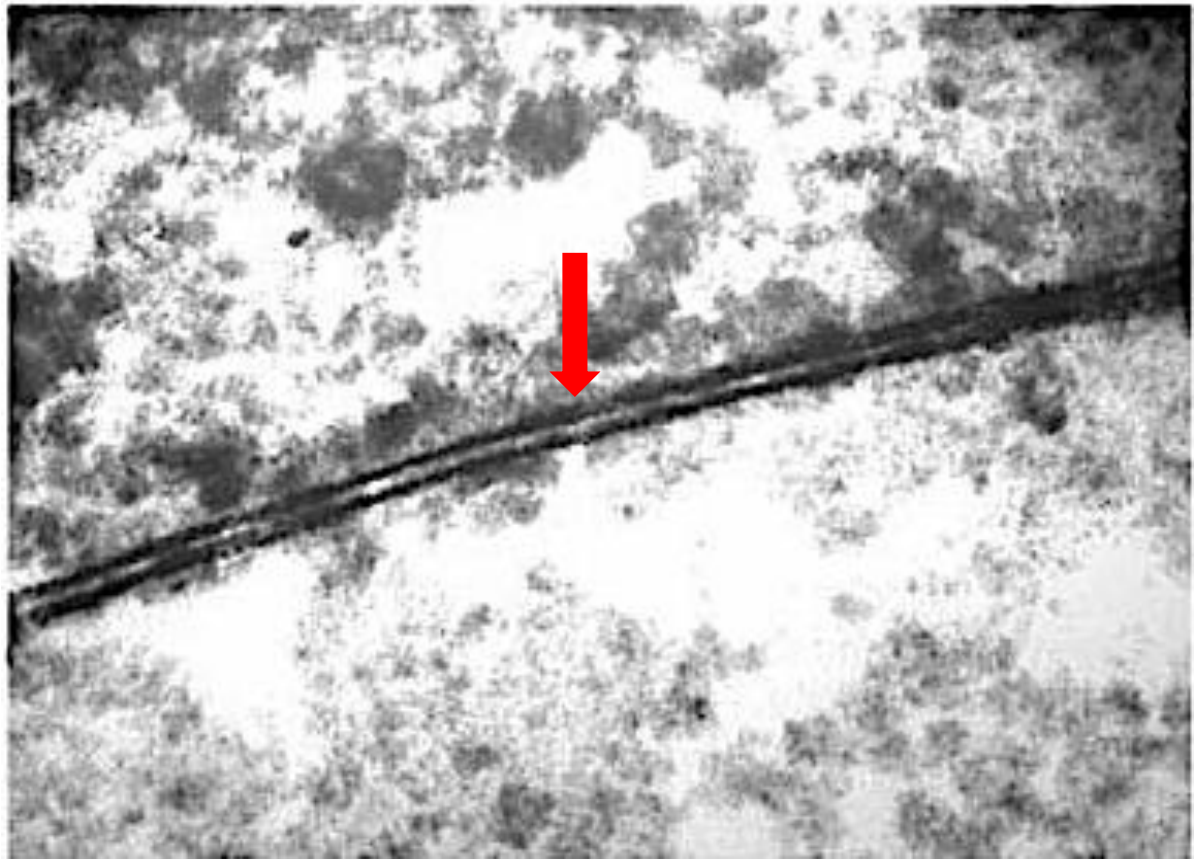


1- Cell Membrane

- **E.M: Trilamellar membrane:**

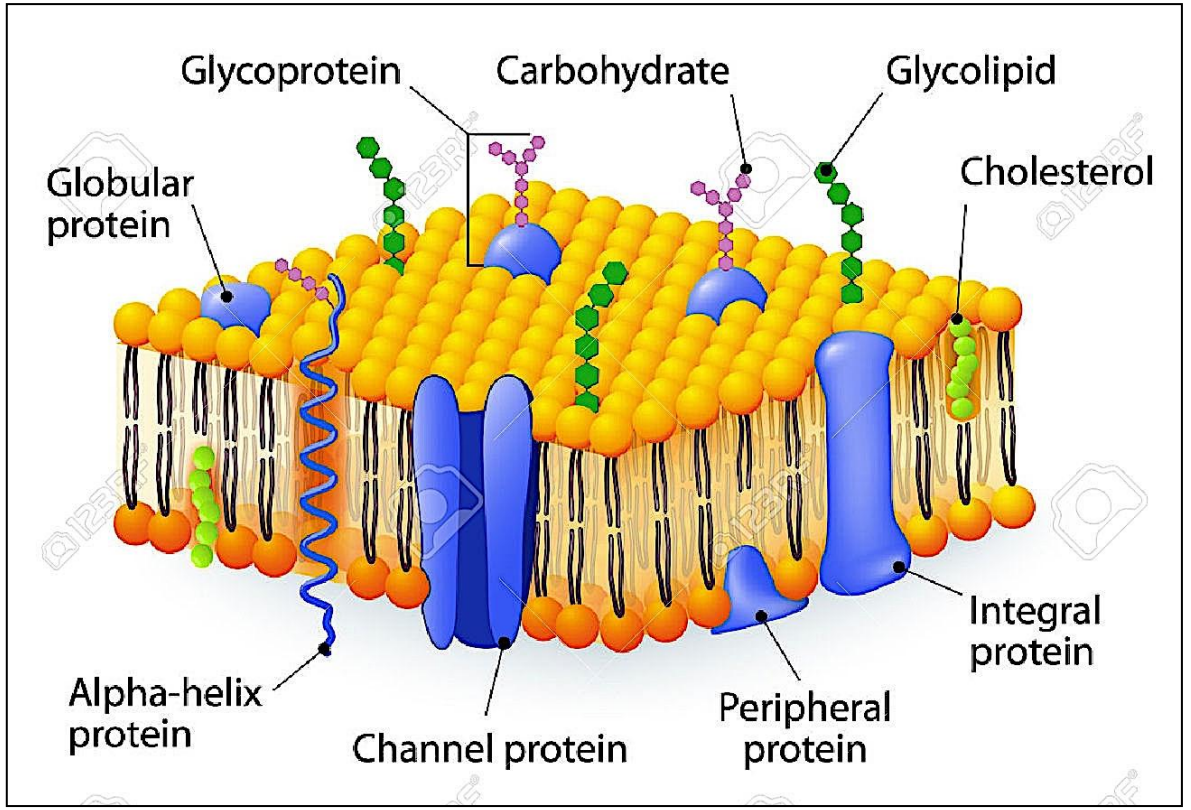
2 dark parallel lines,

separated by light one



Molecular structure of cell membrane

- **Protein Component [60%]**
- **Lipid Component [30%]**
- **Carbohydrate Component [10%]**



Carbohydrate Component [10%]

- **Polysaccharides**

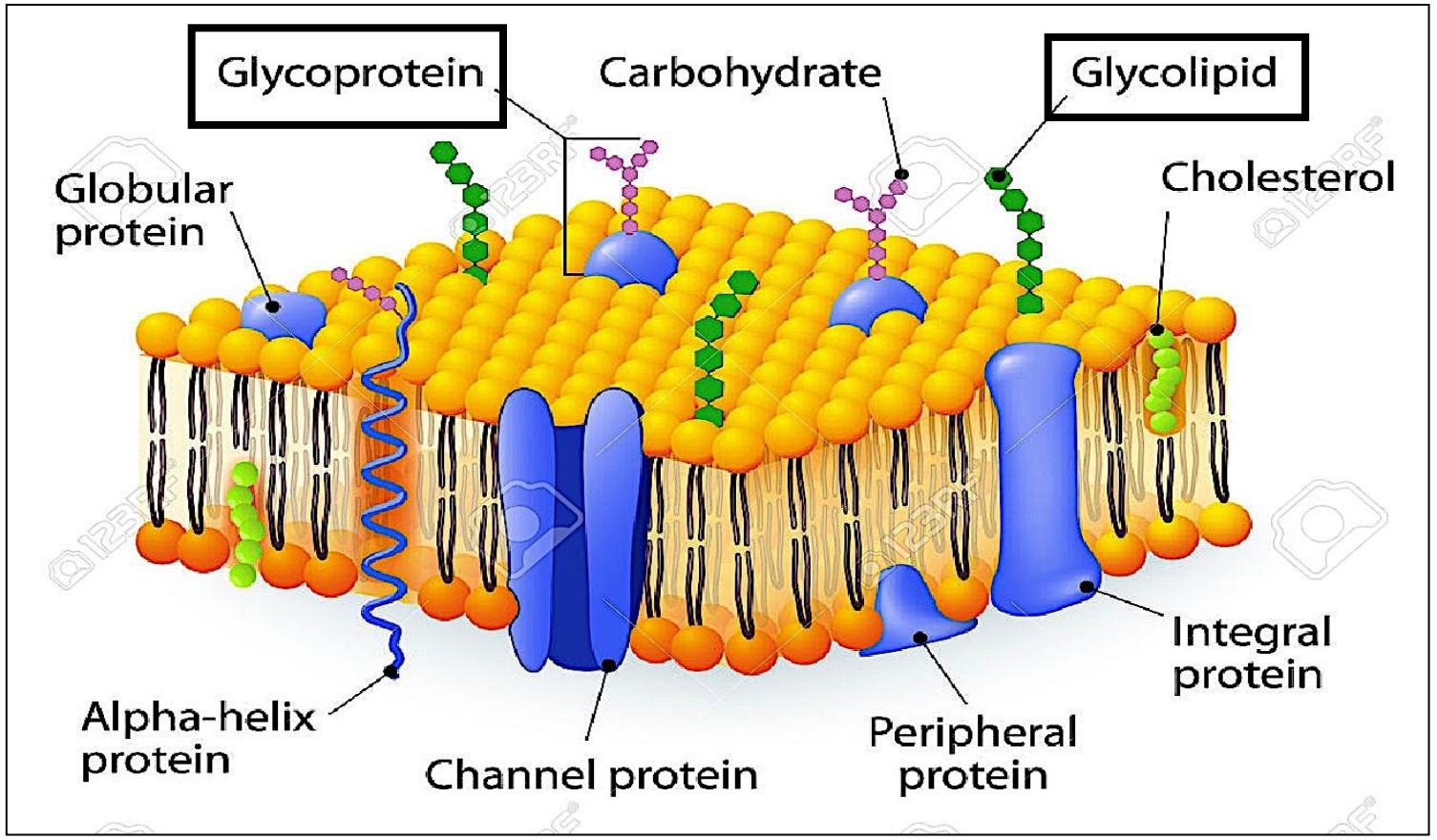
  Either attached to

  protein → **Glycoprotein**

  or to lipid → **Glycolipid**

- **Glycolipid + Glycoprotein**

  → **Cell coat** or **Glycocalyx**



Functions of Cell Membrane

1-Regulate passage of substances in and out of the cell through many mechanisms [Discussed in physiology]: Passive transport, active transport and facilitated transport

2-Functions of cell coat:

- ✓ Mechanical protection of the cell
- ✓ Cell immunity & allergy [through receptors]

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Types of Cytoplasmic Organelles

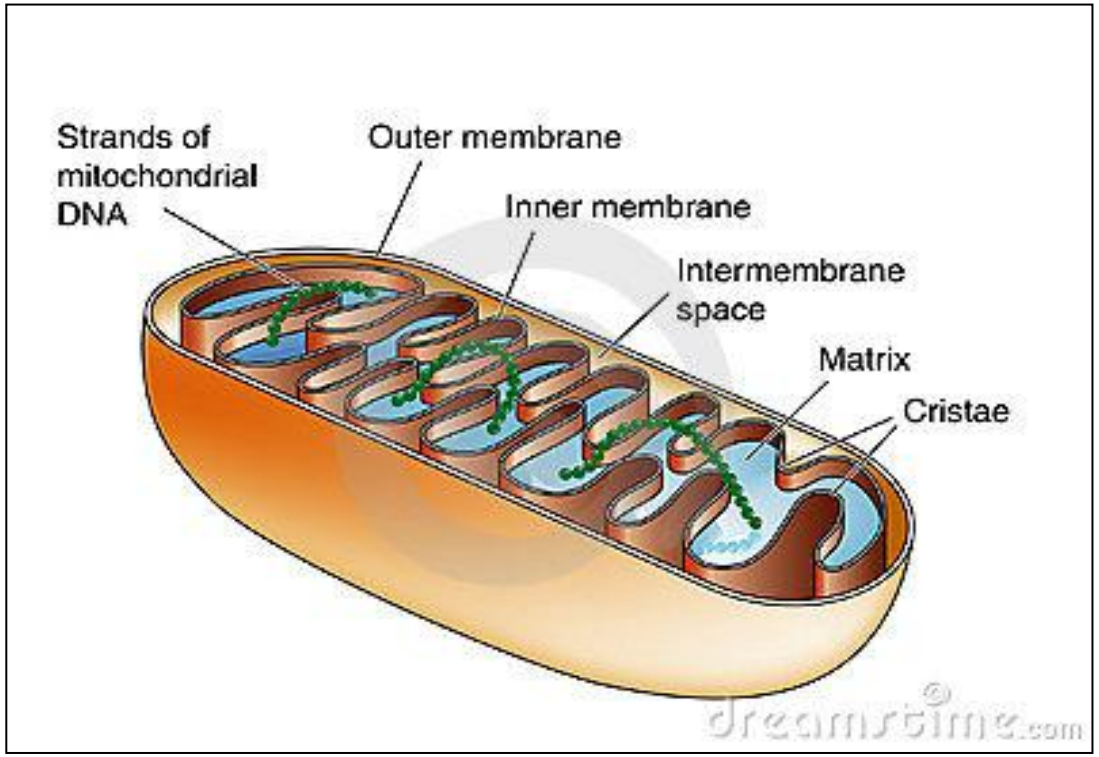
Membranous	Non-membranous
✓ Cell membrane	✓ Ribosomes
✓ Mitochondria	✓ Cytoskeleton: a-Microtubules (Centrioles & Cilia) b-Filaments (Thin, Intermediate & Thick)
✓ Endoplasmic reticulum (rough & smooth)	
✓ Golgi apparatus	
✓ Lysosomes	

Mitochondria

Number:

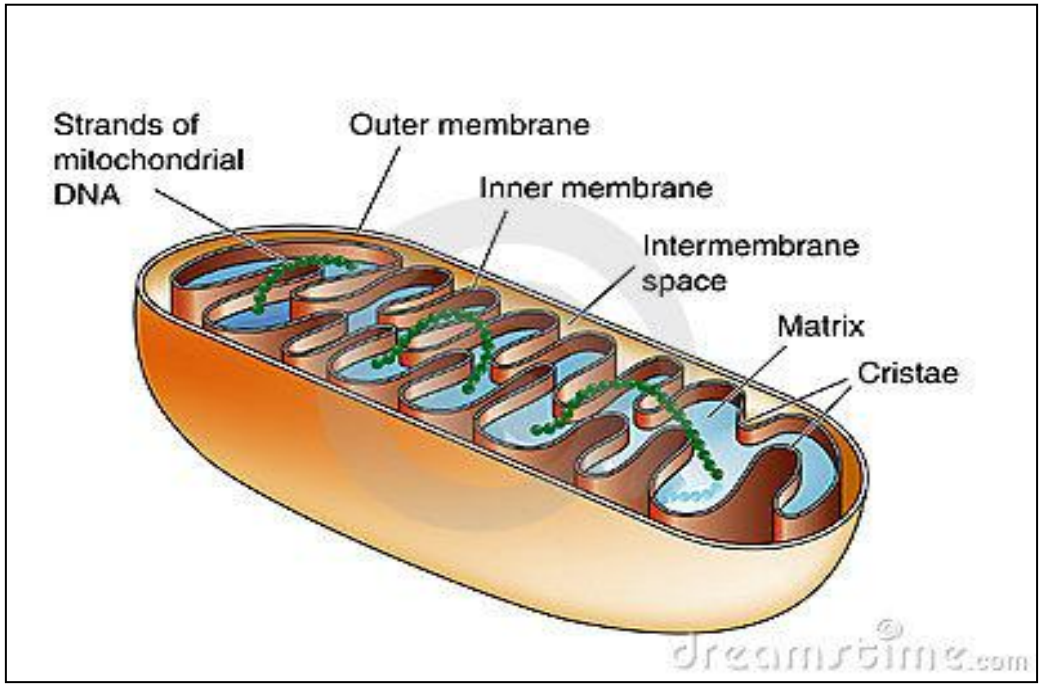
Numerous in active cells

[liver cells contain up to 2000/cell]



Mitochondria

■**E.M:**



✓**Round, oval or elongated vesicles**

✓**Covered with double membranes**

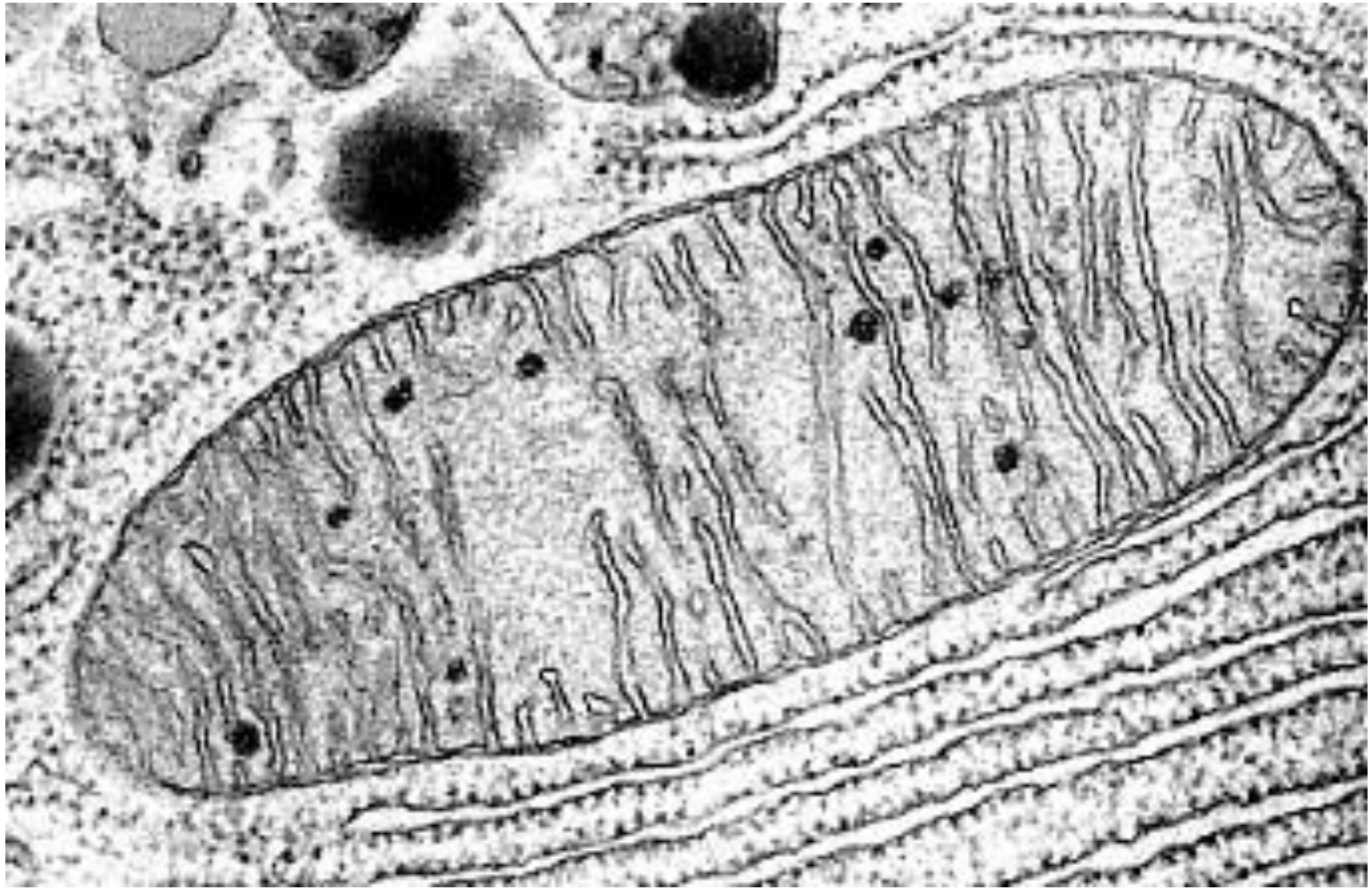
✓**Outer membrane is smooth**

✓**Inner membrane form folds [Cristae]**

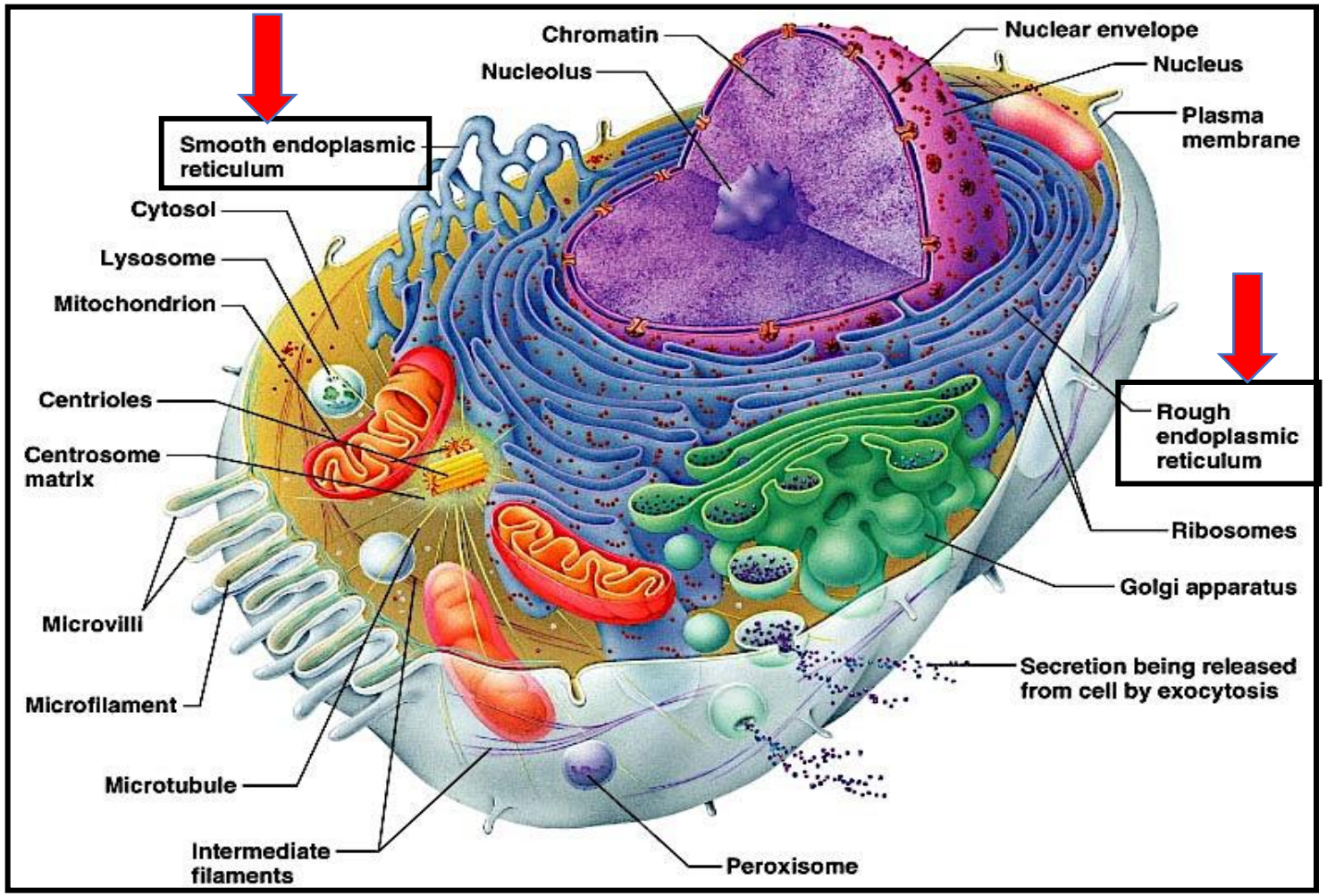
✓**Matrix fills the internal cavity, contains lipid, protein, carbohydrate, Ca, Mg, DNA, RNA & Oxidative enzymes.**

■**Function: formation of ATP which is the main source of energy (they are the powerhouses of living cells)**

Mitochondria [E.M]



The Cell


Chromatin
Nuclear envelope
Nucleolus
Nucleus
Plasma membrane
Smooth endoplasmic reticulum
Cytosol
Lysosome
Mitochondrion
Centrioles
Centrosome matrix
Rough endoplasmic reticulum
Ribosomes
Golgi apparatus
Microvilli
Secretion being released from cell by exocytosis
Microfilament
Microtubule
Intermediate filaments
Peroxisome

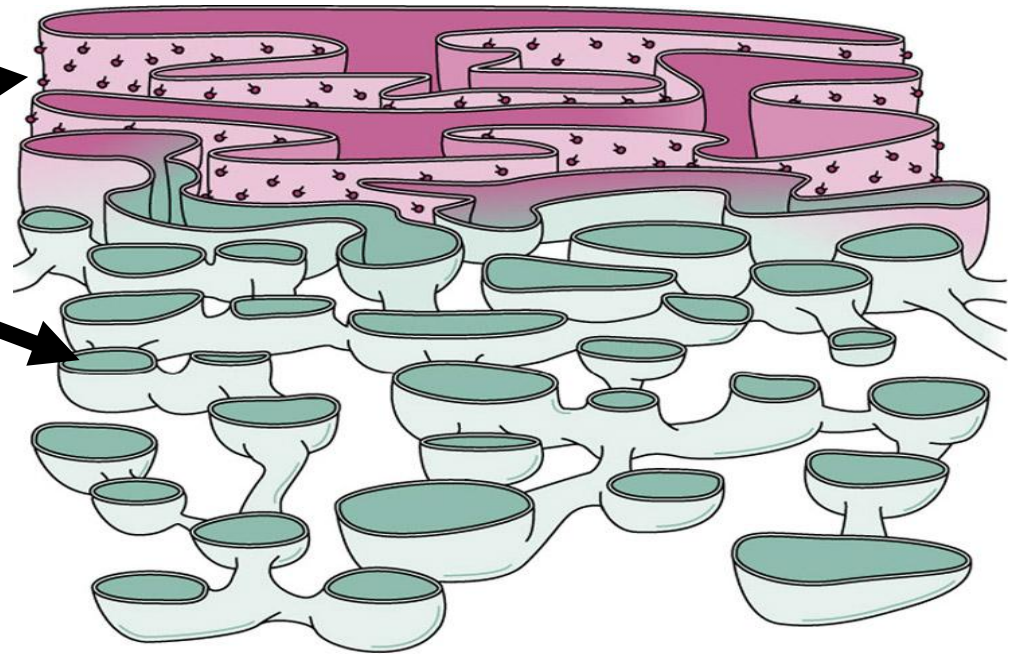
Endoplasmic Reticulum

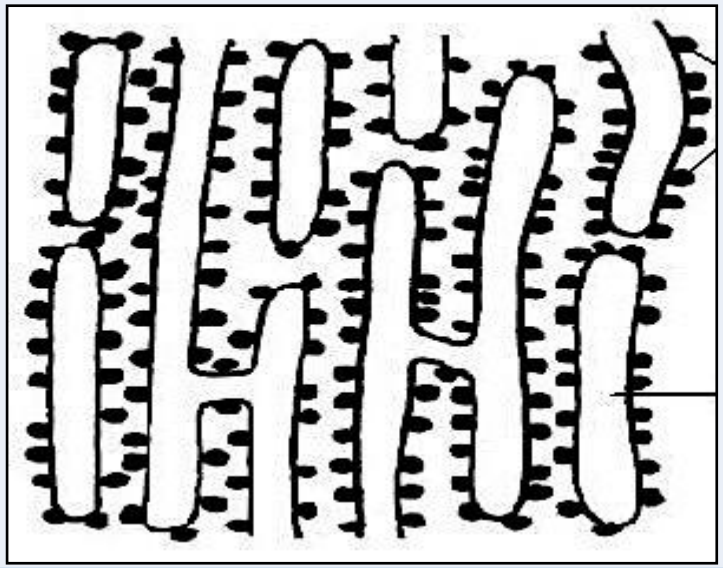
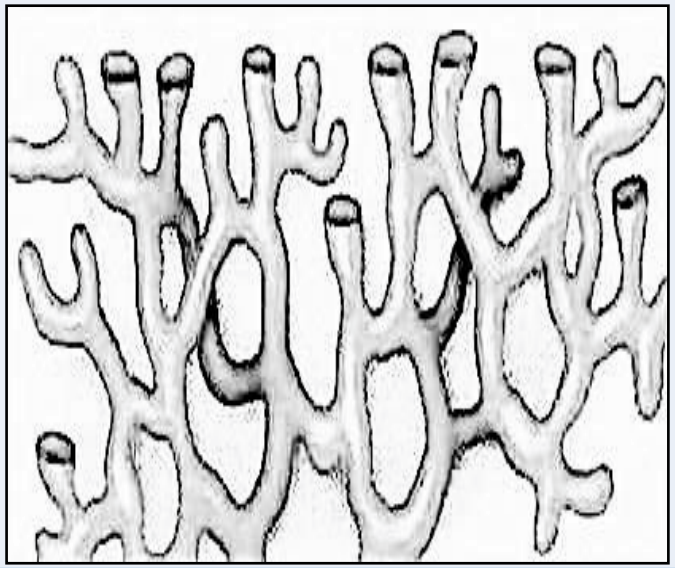
■**Def:** Network [reticulum] of intercommunicating channels in cytoplasm.

■**Types:**

A-**Rough** (with ribosomes)

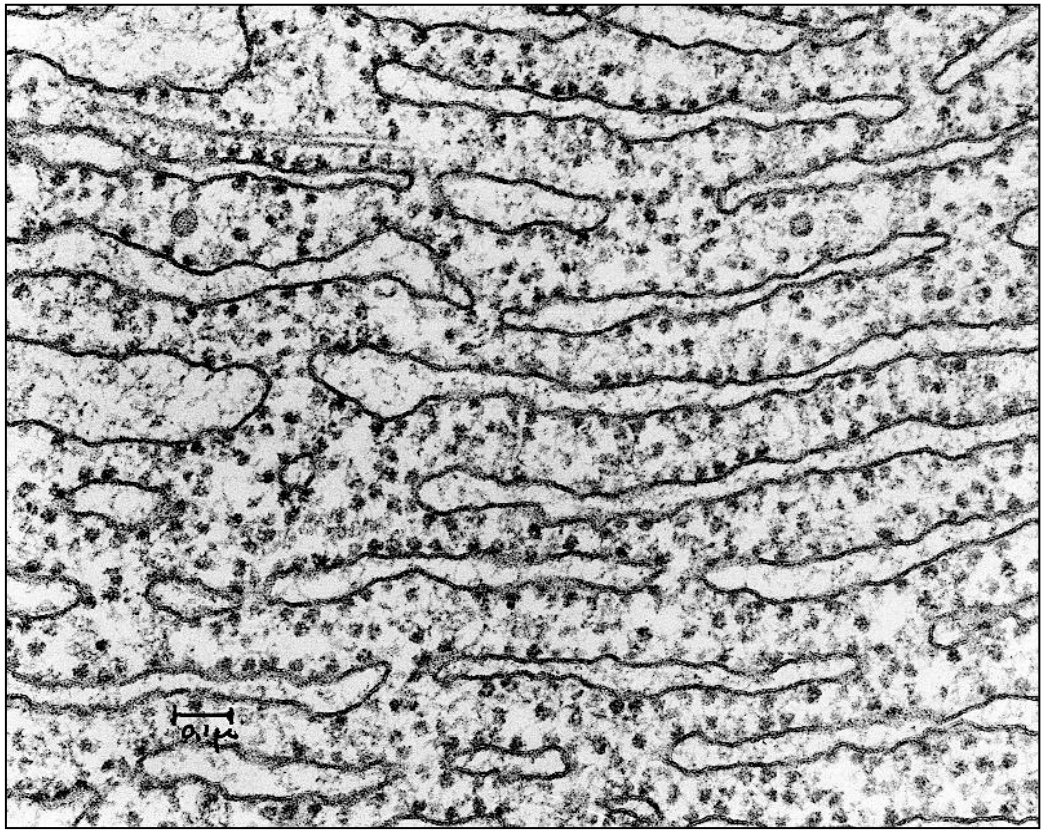
B-**Smooth** (without ribosomes)



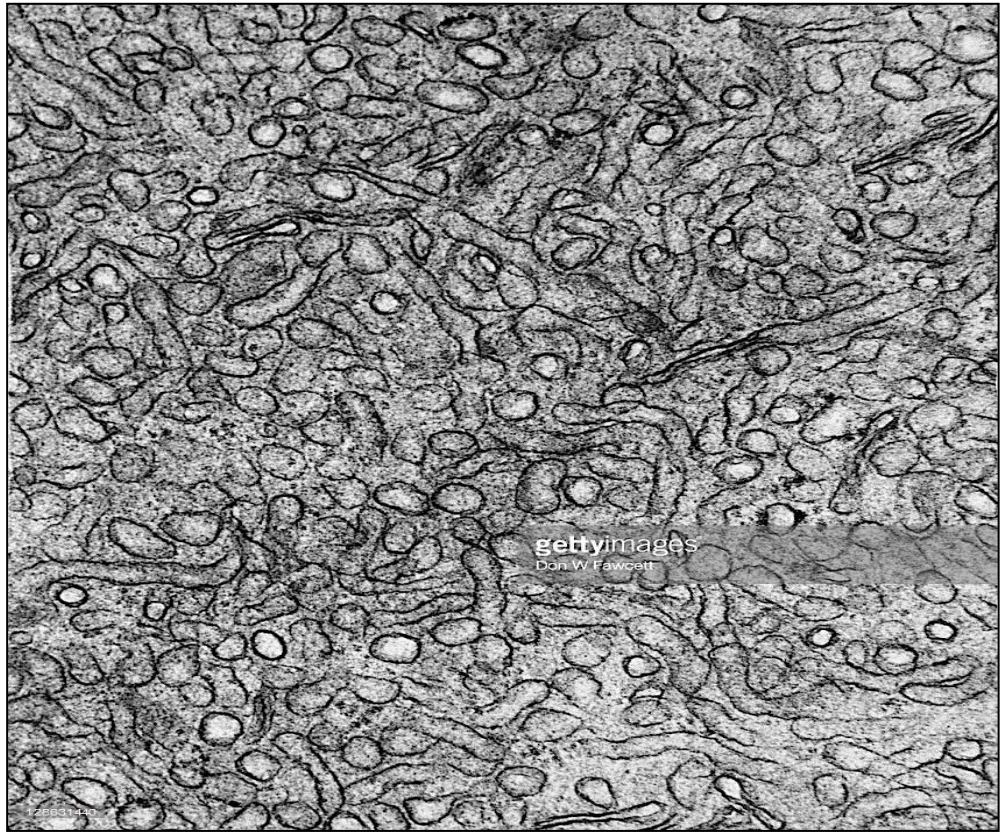
Types	rER	sER
		
Site	✓ **Protein** forming cells: Plasma cells, Osteoblasts & Pancreas	✓ **Lipid** forming cells: Liver ✓ **Endocrine glands**: Suprarenal gland

E.M	Attached ribosomes → Rough outer surface	No ribosomes → Smooth outer surface
Function	Synthesis of protein by ribosomes → transfer of protein to Golgi apparatus	1-Synthesis of Lipids & steroid hormones 3-Storage of glycogen [in liver & muscles] 4-Storage and pump of Ca → muscle contraction 5-Detoxification of alcohol & some drugs [in liver]

EM picture of rER



EM picture of sER

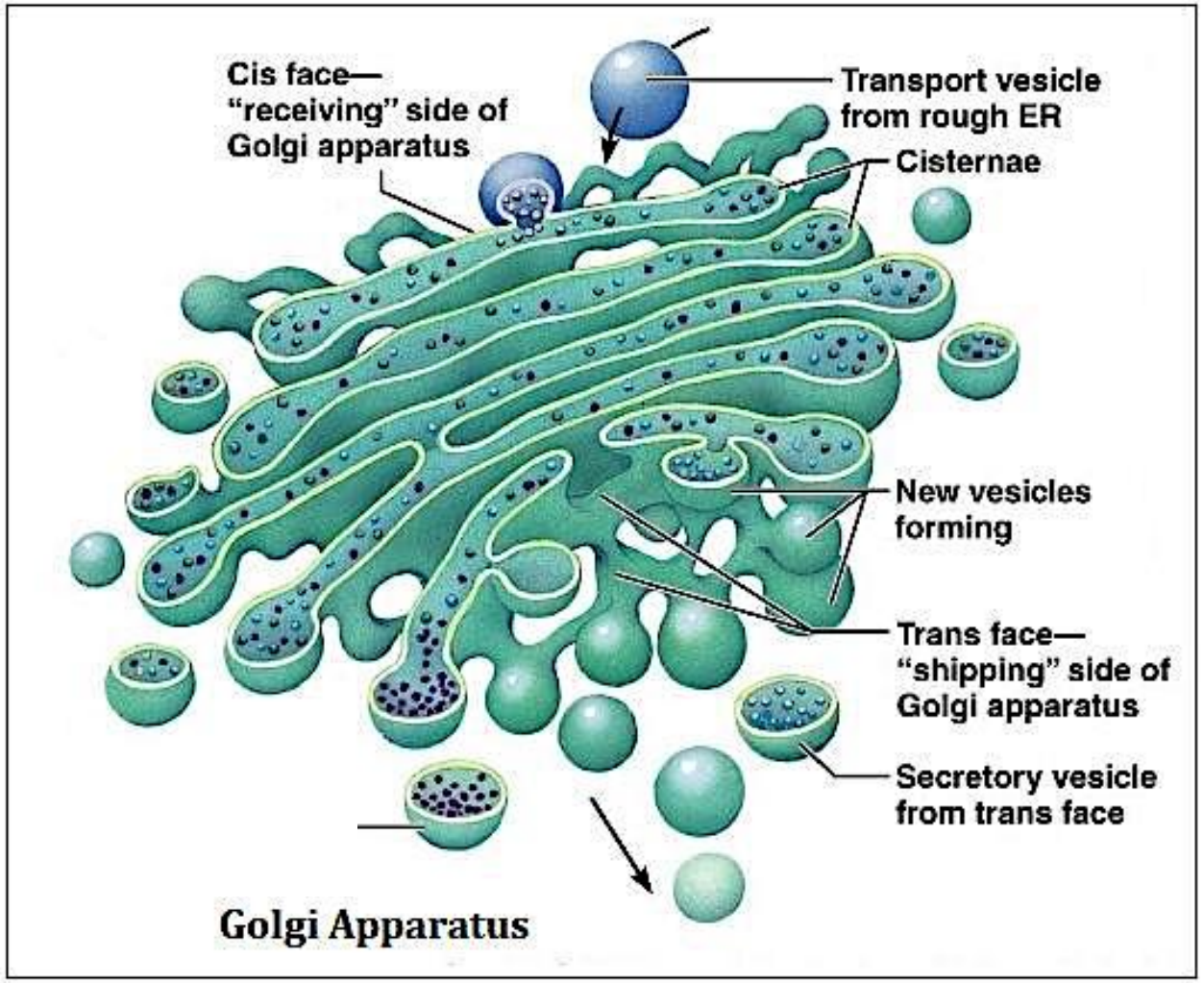


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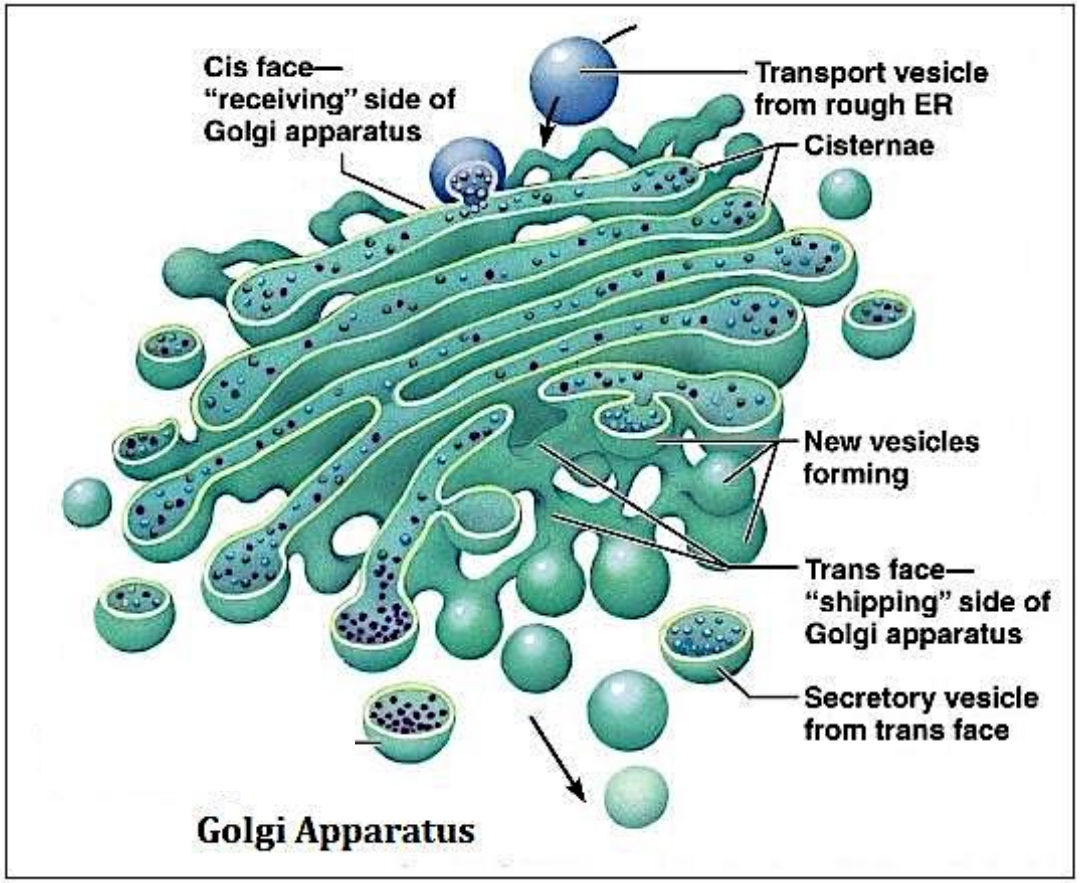
Golgi Apparatus



Golgi Apparatus

■Def:

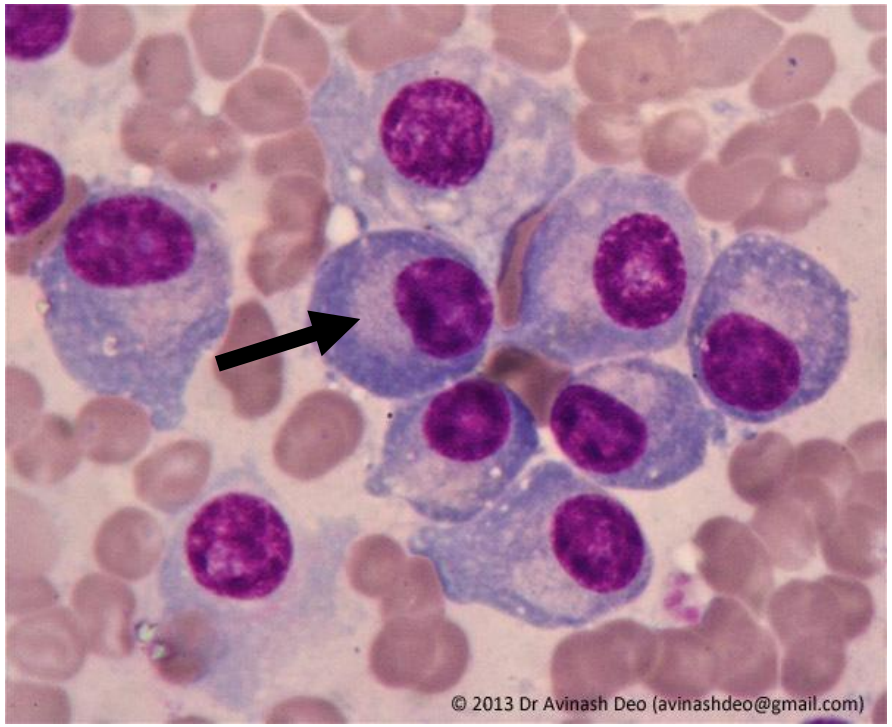
- ✓ The **secretory system** of the cell [Named after histologist **Camillo Golgi,** discovered in 1898]
- ✓ Well developed in secretory cells



Golgi Apparatus

■L.M:

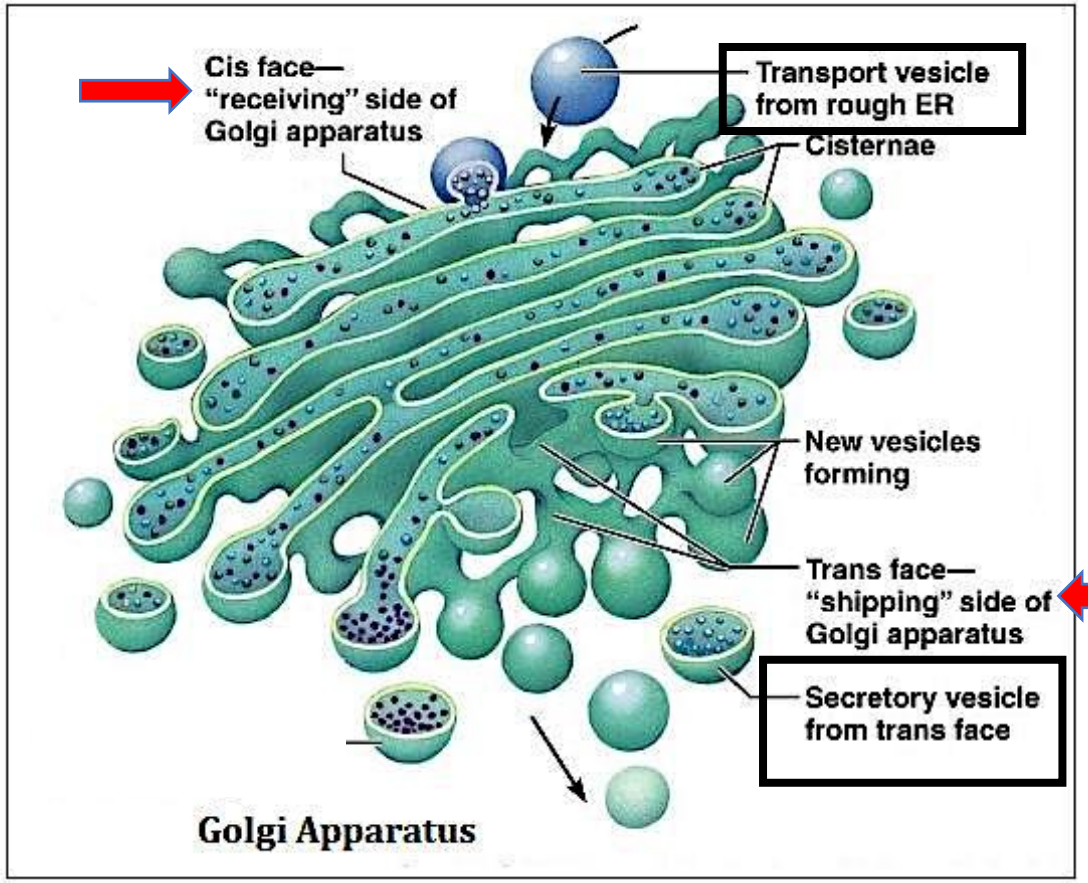
✓**Not** stainable with H&E (dissolves during preparation due to **high lipid** content in its membranes) → Gives **pale area** in cytoplasm called **"negative Golgi image"** or **"Golgi ghost"**



Golgi Apparatus

■E.M:

- ✓Many flat curved membranous saccules
- ✓Stacked over each other



Golgi Apparatus

■**E.M:**

✓**Each saccule has two faces:**

a-Immature (Cis) face:

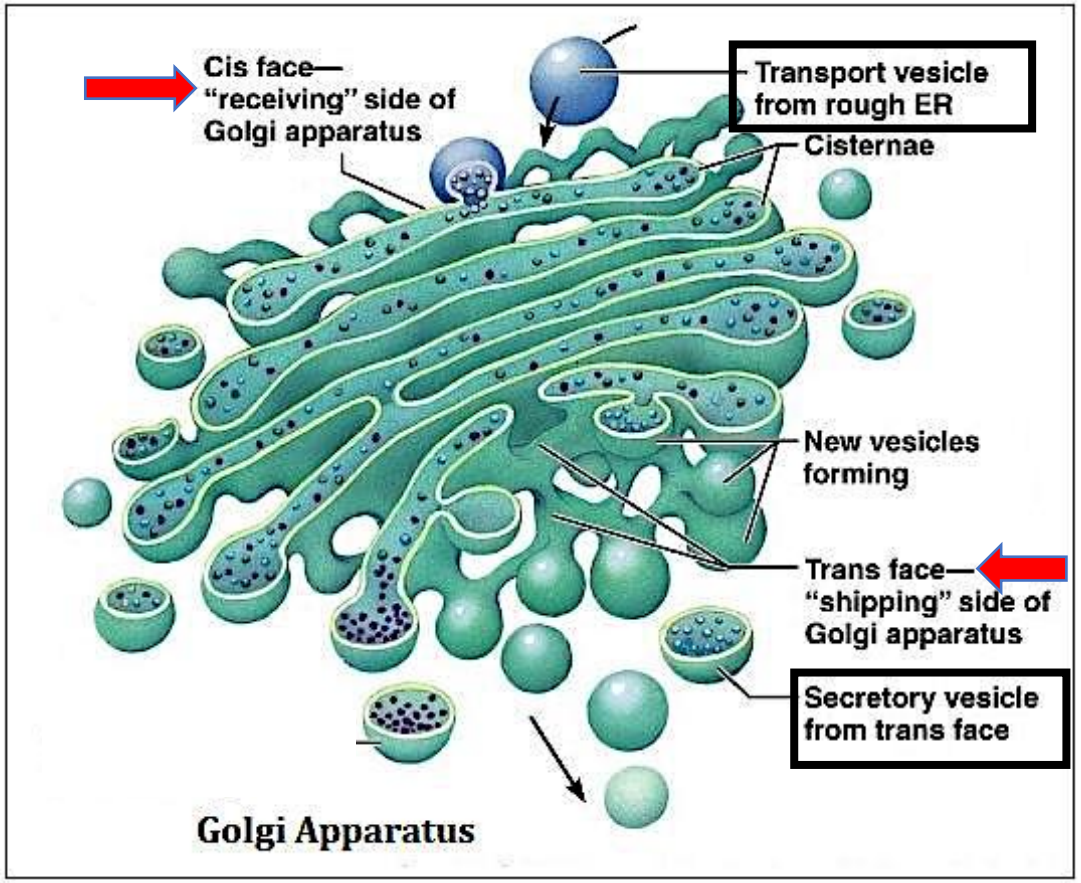
It receives transport vesicles from rER

b-Mature (Trans) face:

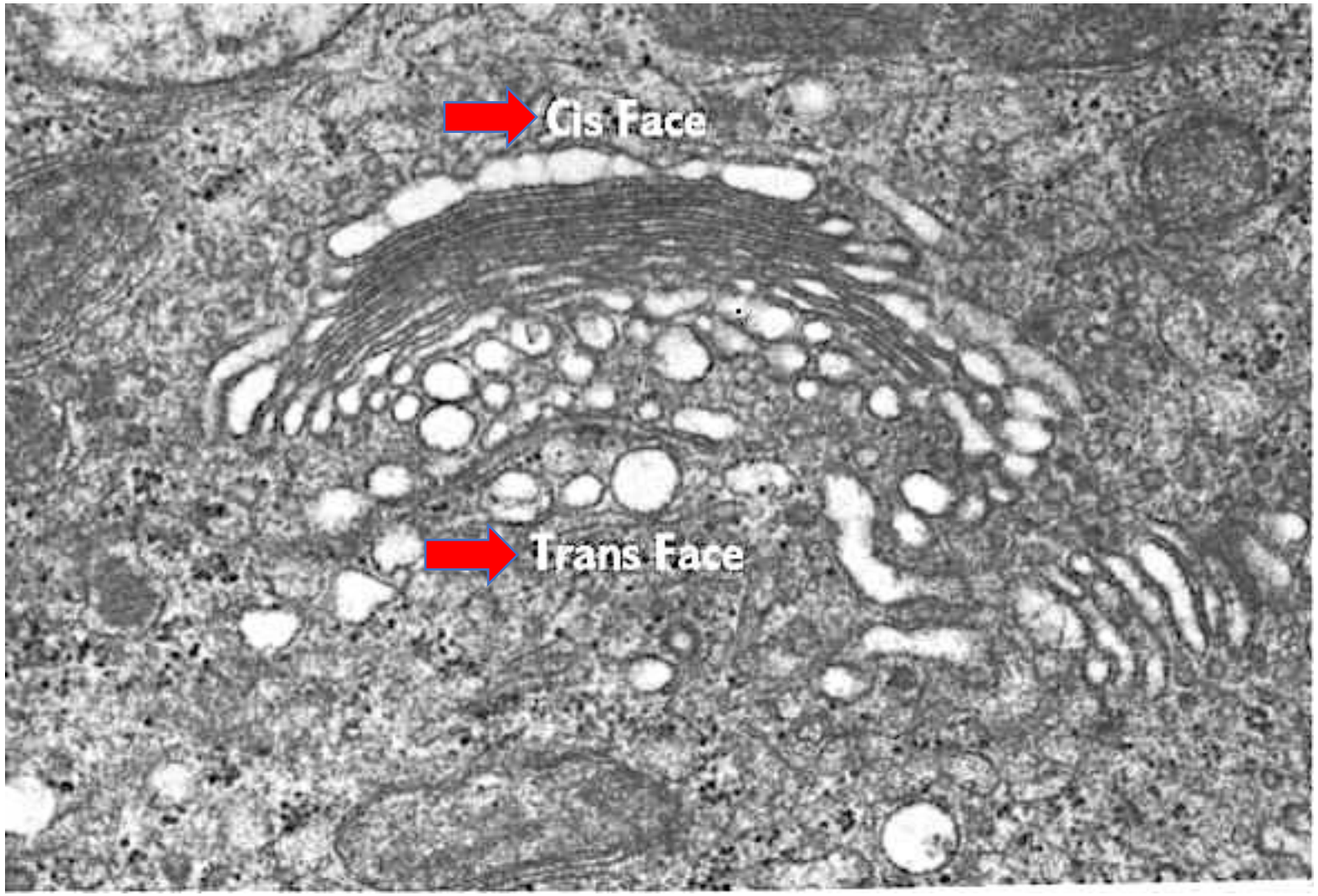
Is the "shipping area" that gives two types of vesicles:

✓**Secretory vesicles for exocytosis**

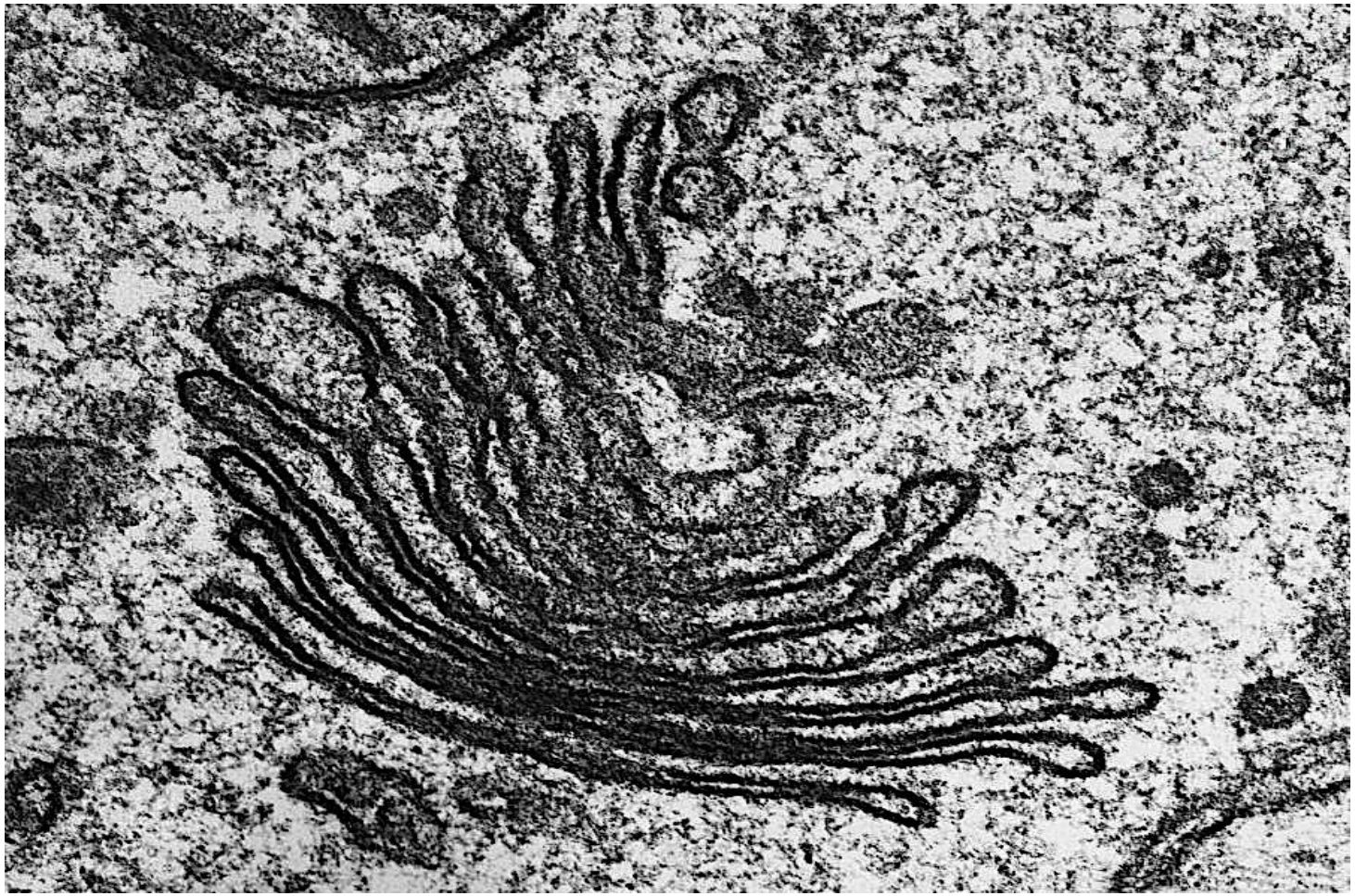
✓**Lysosomes**



E.M picture of Golgi Apparatus



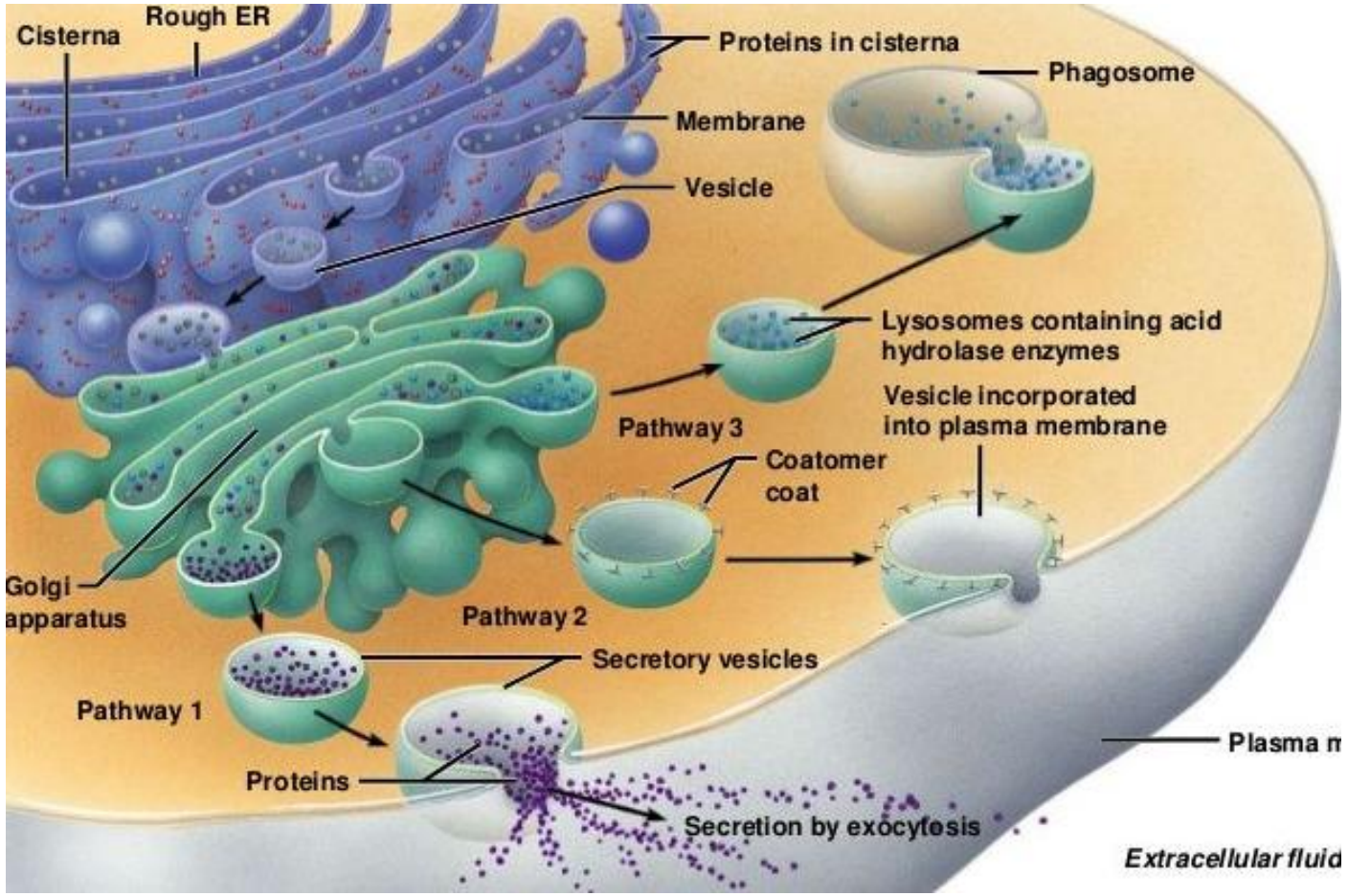
E.M picture of Golgi Apparatus



Functions of Golgi Apparatus

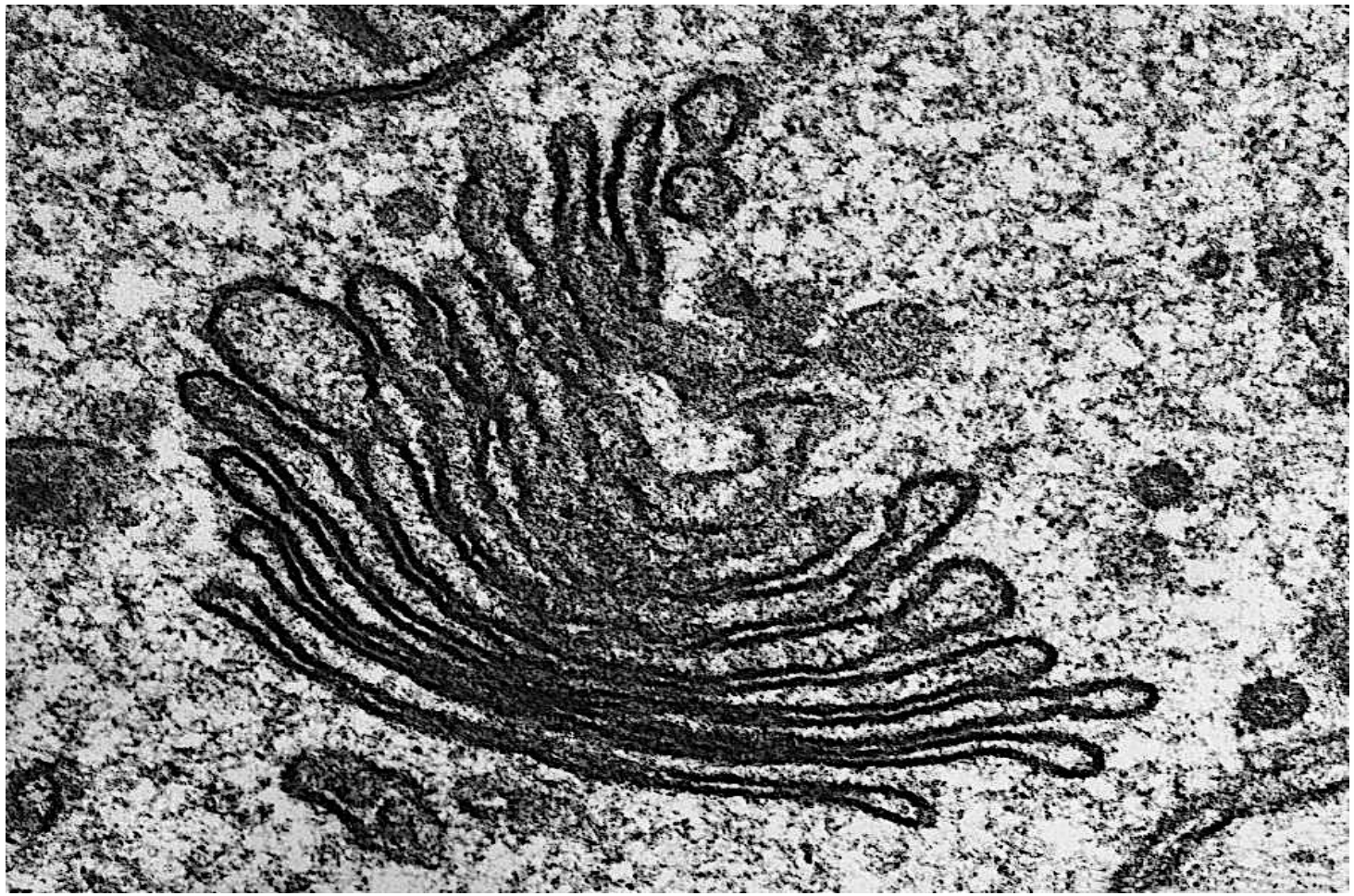
Completes the function of rER by:

- **Modification** of **proteins** formed by rER → **Discharged** as secretions
- **Formation of lysosomes** & isolation of their enzymes



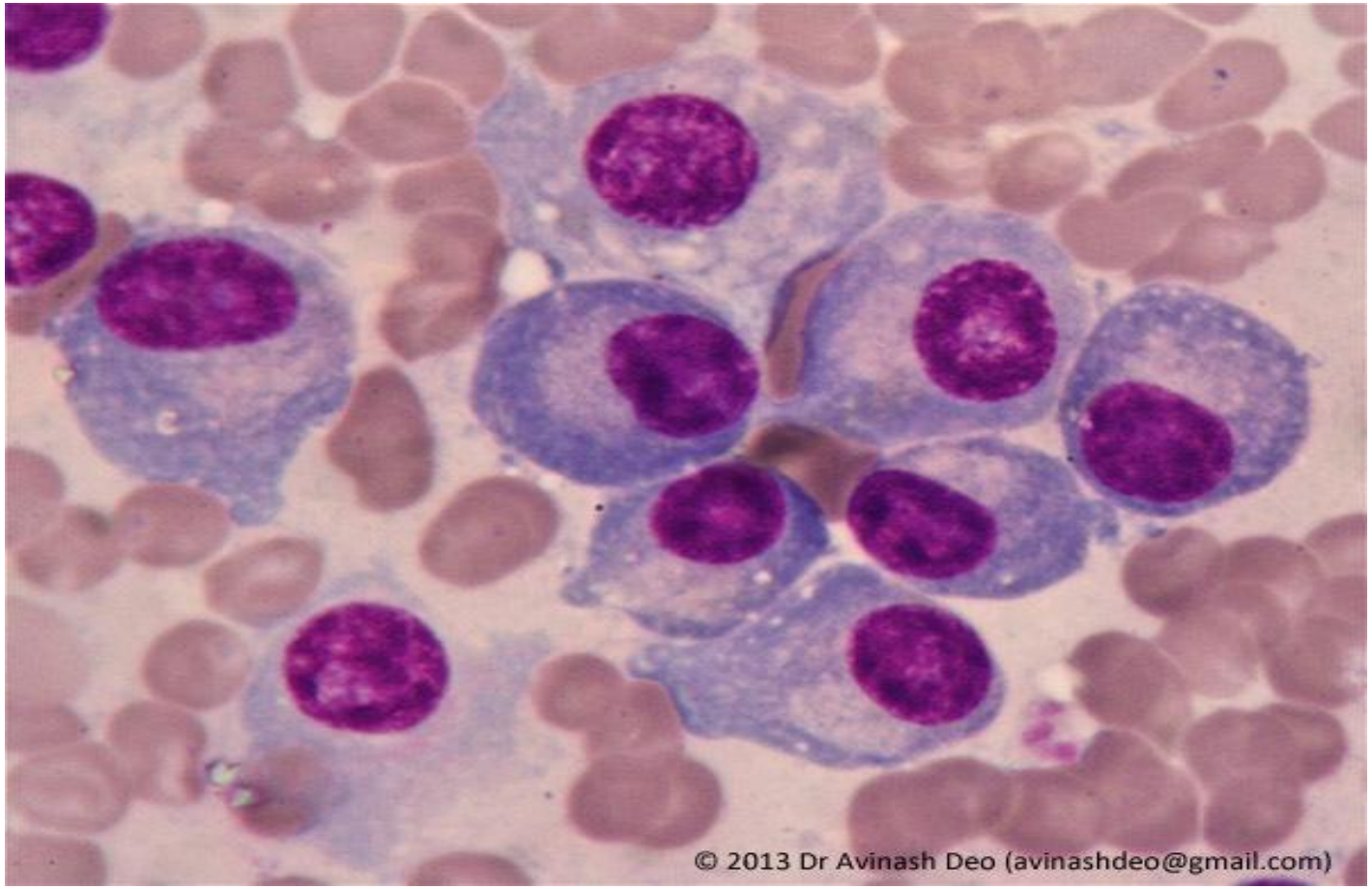
Practical slides of this session

?



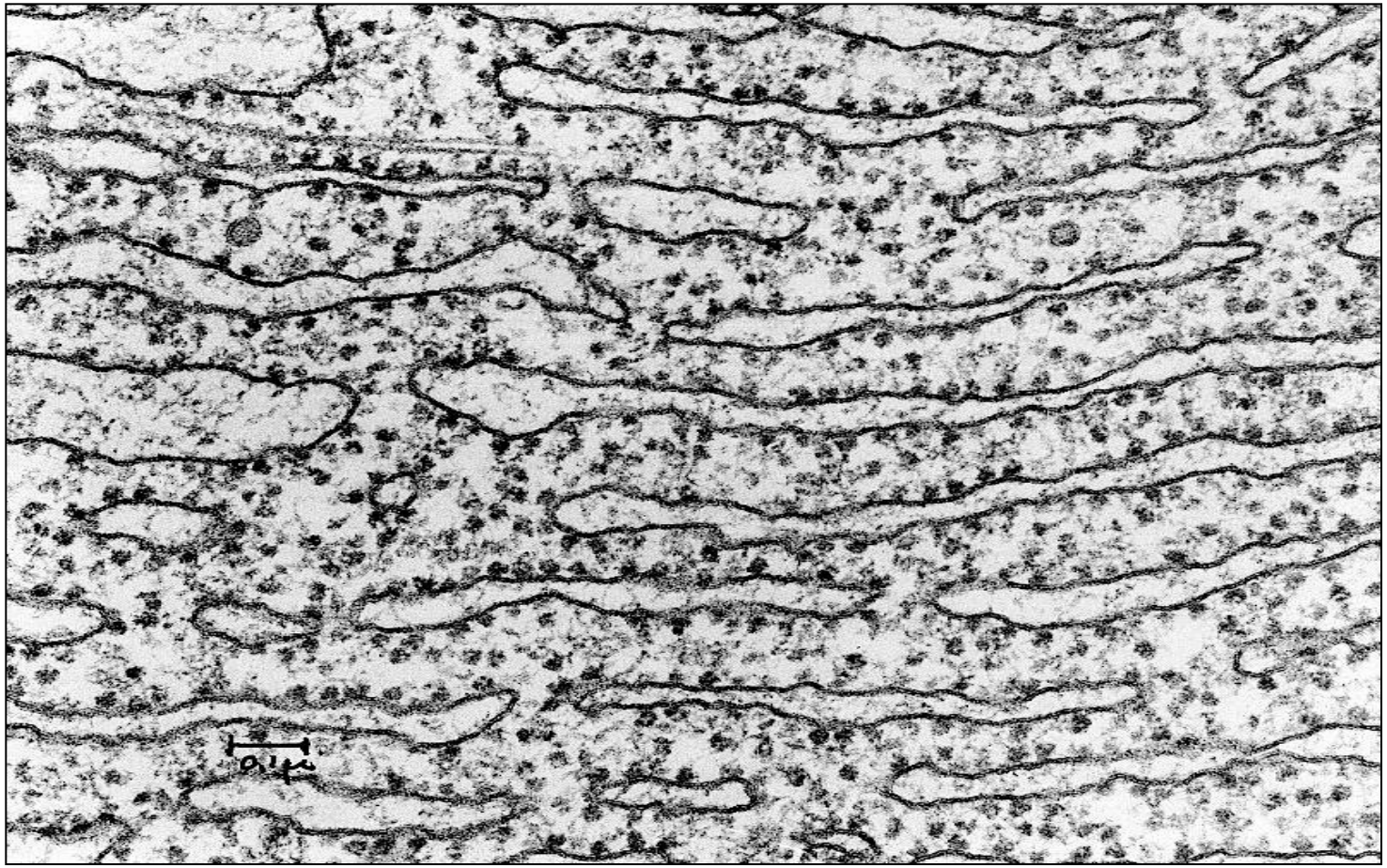
EM picture of Golgi Apparatus

?

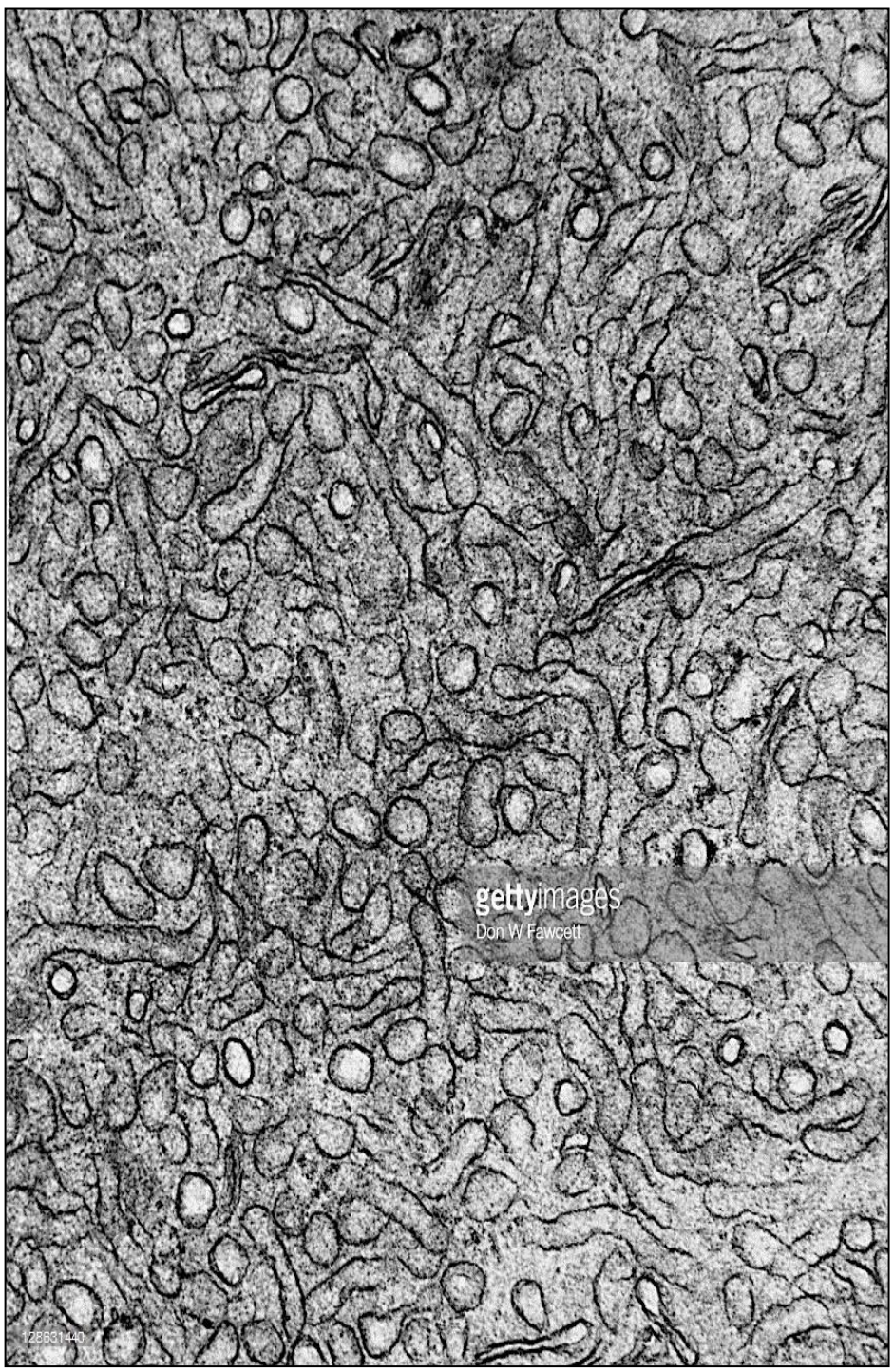

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LM picture of negative Golgi Image

?

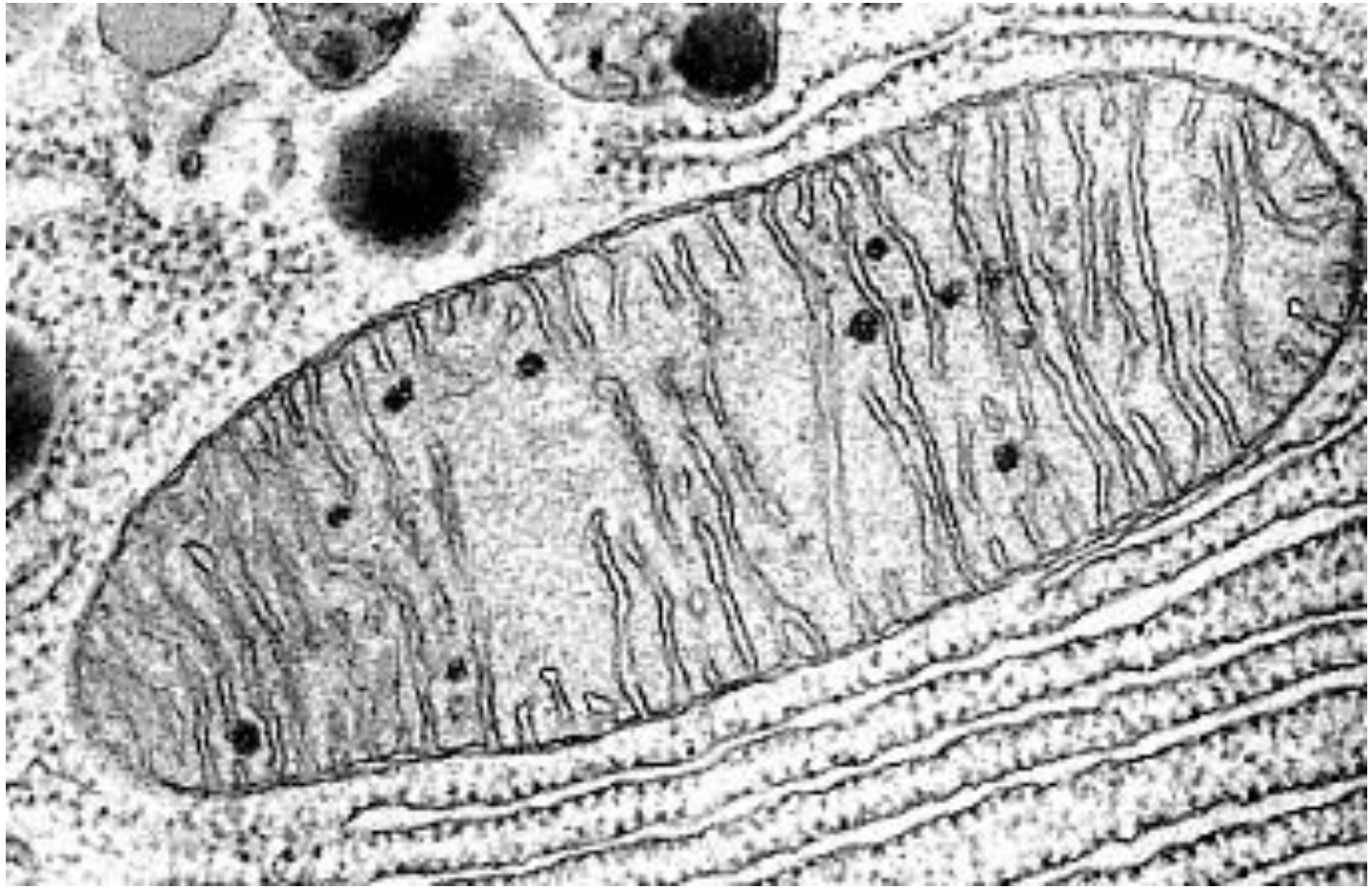


EM picture of rough ER


gettyimages
Don W Fawcett

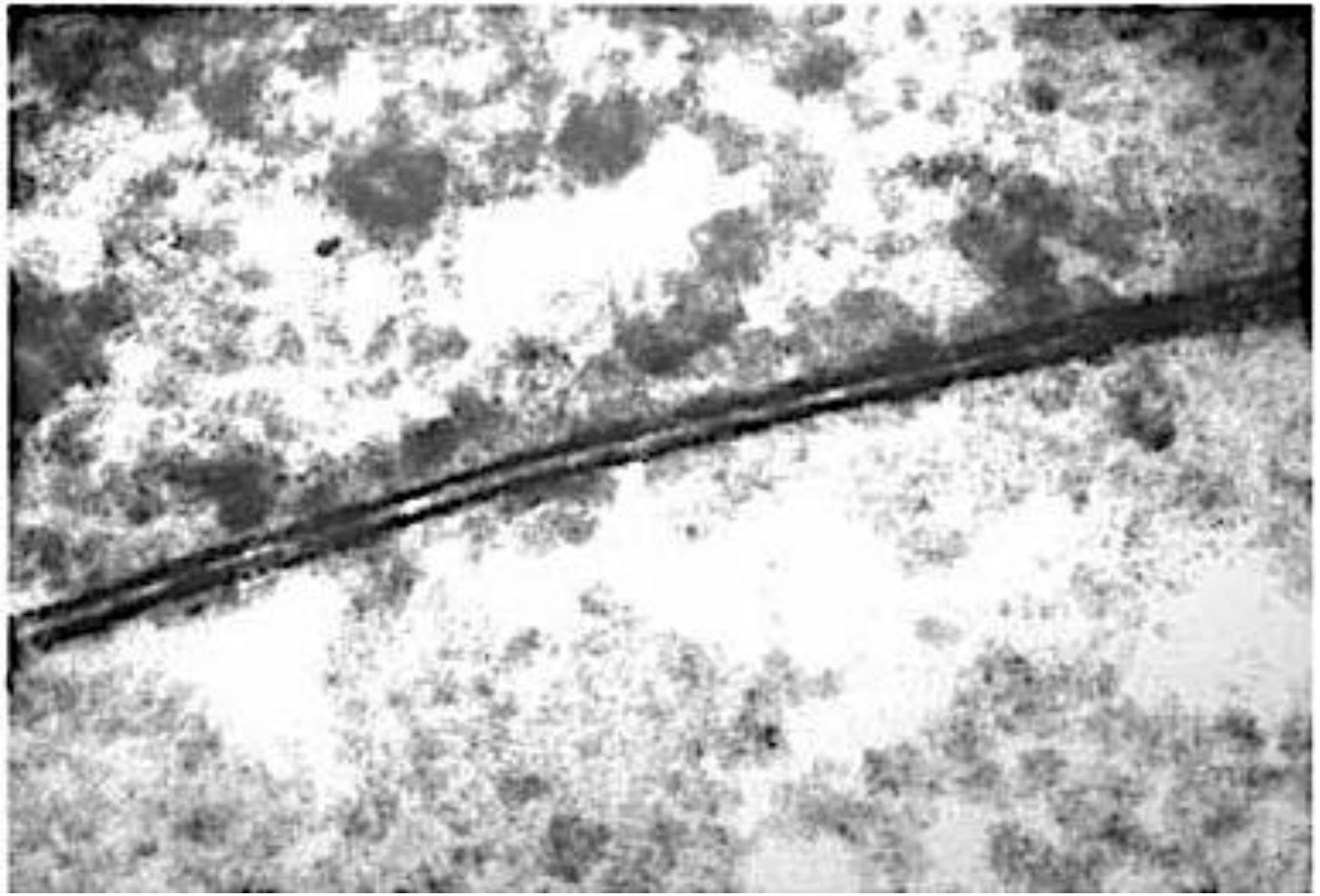
EM picture of smooth ER

?



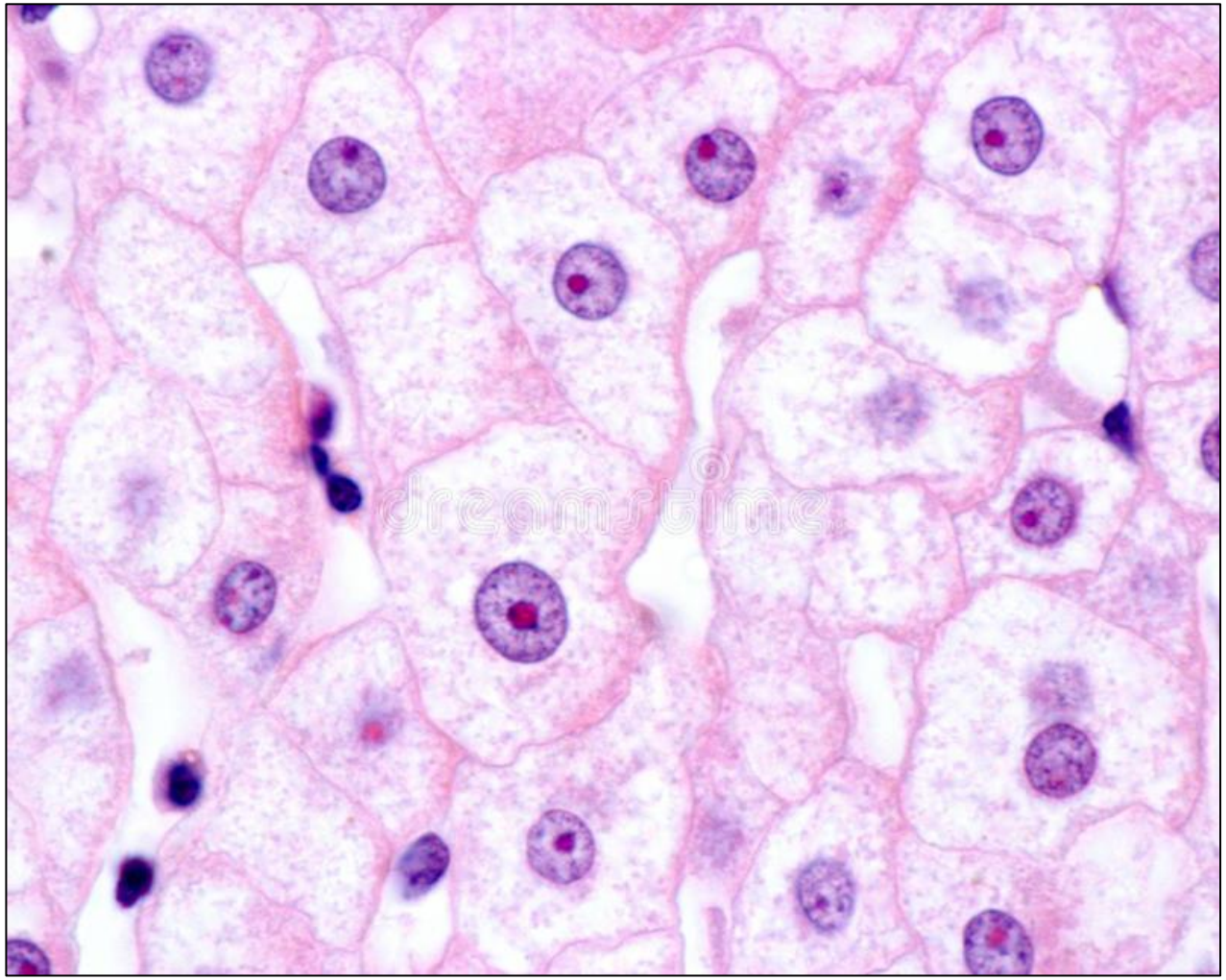
EM picture of mitochondrion

?



EM picture of cell membrane

?



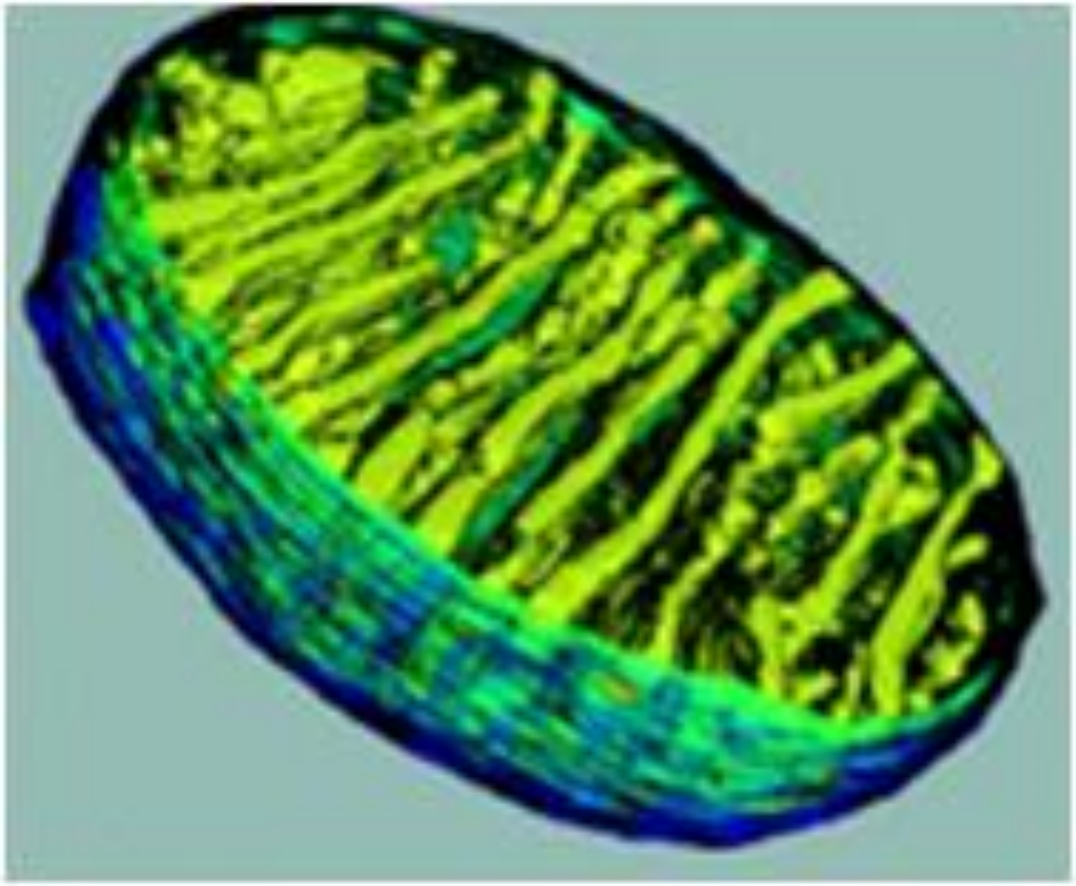
LM picture of a cells

?



Scanning EM picture of mitochondria and ER

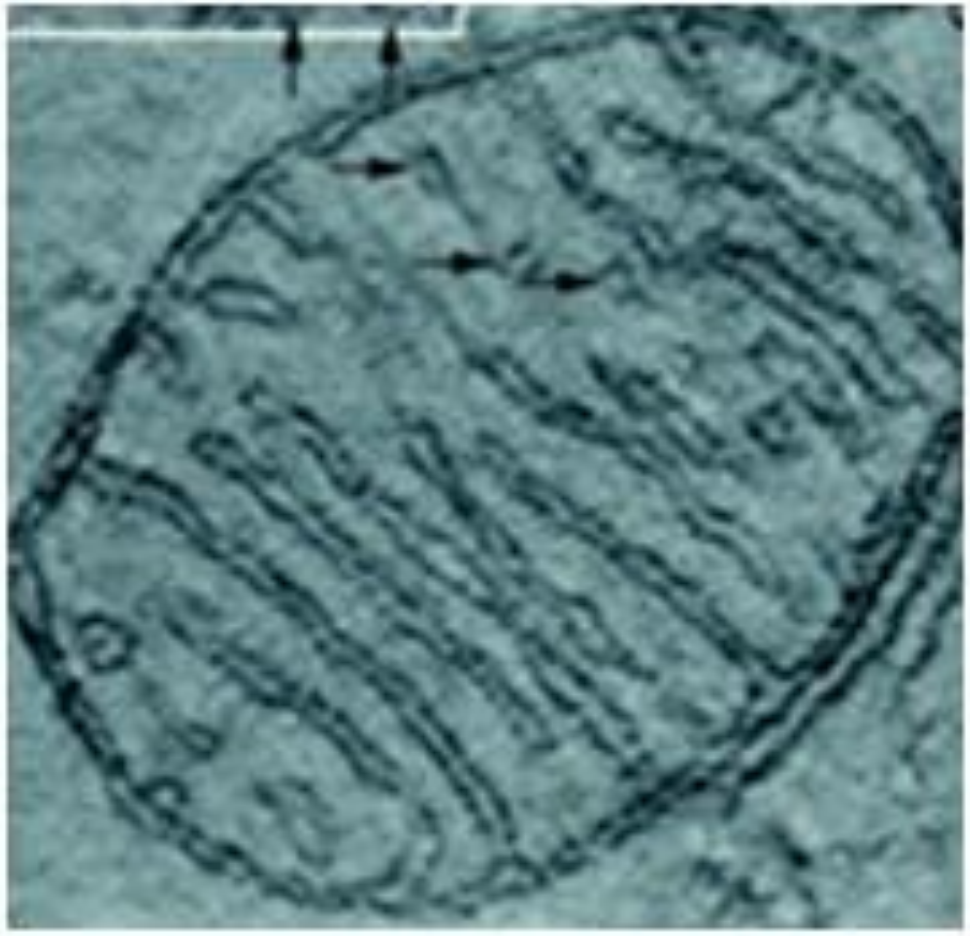
?



Scanning EM picture of a mitochondrion

?

- .



EM picture of a mitochondrion

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THANK YOU

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El Tur Campus : Sahely Road Ras suder Campus : Obour Street Sharm El Sheikh Campus : El Nour District